

Fundamentals and Principles of Terahertz (THz) Spectroscopy

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Last modified on: 12/06/2026

1.) Introduction:

- 1.a.) What is THz spectroscopy?**
- 1.b.) The THz gap: technological and experimental challenges**
- 1.c.) History, growing interest and current applications**

2.) Linear THz Spectroscopy:

- 2.a.) Fourier-transform infrared (FTIR) spectroscopy**
- 2.b.) How to generate and detect THz radiation**
- 2.c.) Examples from literature**

3.) Nonlinear THz Spectroscopy:

- 3.a.) From probing to driving: entering the nonlinear regime**
- 3.b.) Strong-field THz sources**
- 3.c.) Field-resolved THz detection**
- 3.d.) THz time-domain spectroscopy (THz-TDS)**
- 3.e.) Optical-pump-THz-probe spectroscopy and examples**

4.) Summary:

- 4.a.) Take-home messages**
- 4.b.) Guiding questions for reading a THz spectroscopy paper**
- 4.c.) Further reading**

Literature Recommendation



Beginner-friendly

A well-written, lively and understandable overview of THz techniques.

Main reference for this lecture

Today's lecture is strongly inspired by the structure and explanations of this book.

More than we can cover today...

The book also discussed additional aspects/ applications beyond this 90 minutes lecture.

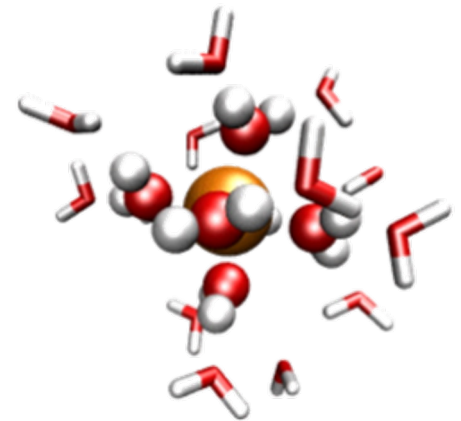
Feedback is very welcome! If you notice any mistakes or unclear points in the following slides, please let me know via email 😊

What is THz Spectroscopy?

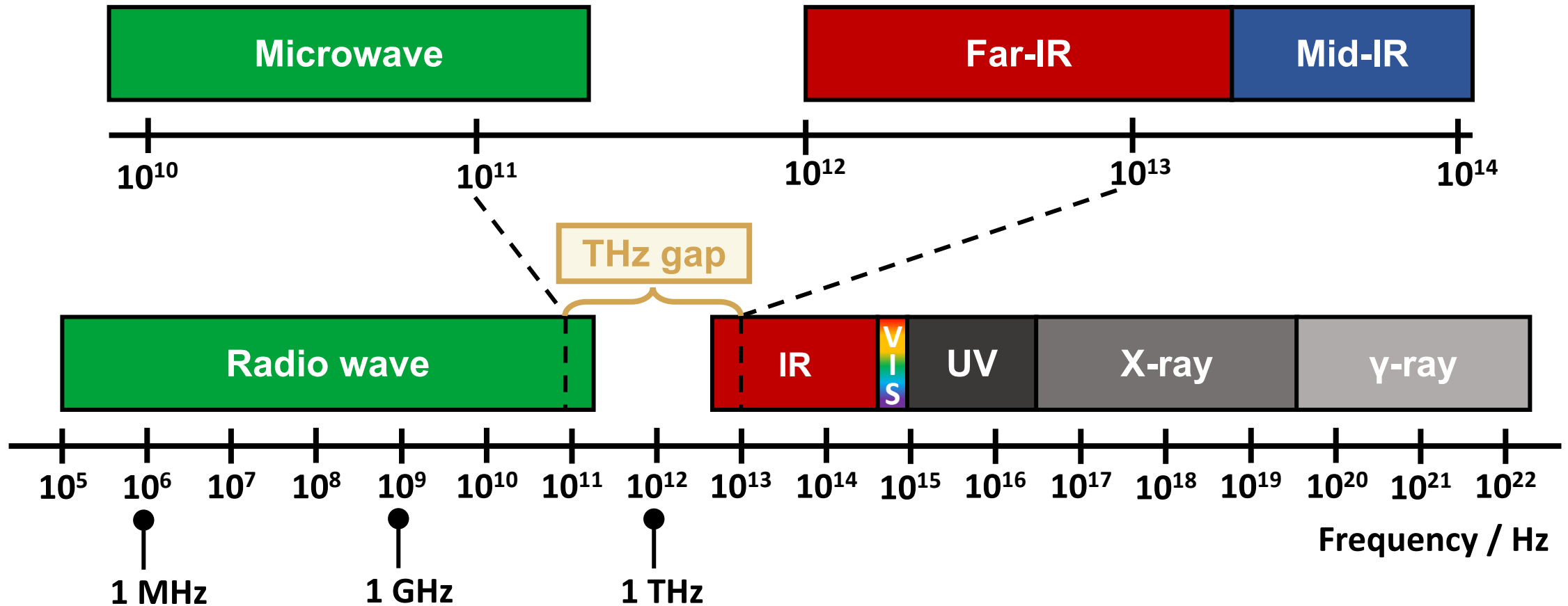
Definition: THz spectroscopy probes the interaction of matter with electromagnetic radiation in the THz range to extract **information about low-energy excitations** (e.g., rotational and vibrational transitions, collective phonon modes in solids, intermolecular interactions, ...).

Historically, this field was called **“far-infrared” spectroscopy** (sometimes still used today!)

Typical range: $\approx 0.1 - 10$ THz (not strictly defined)

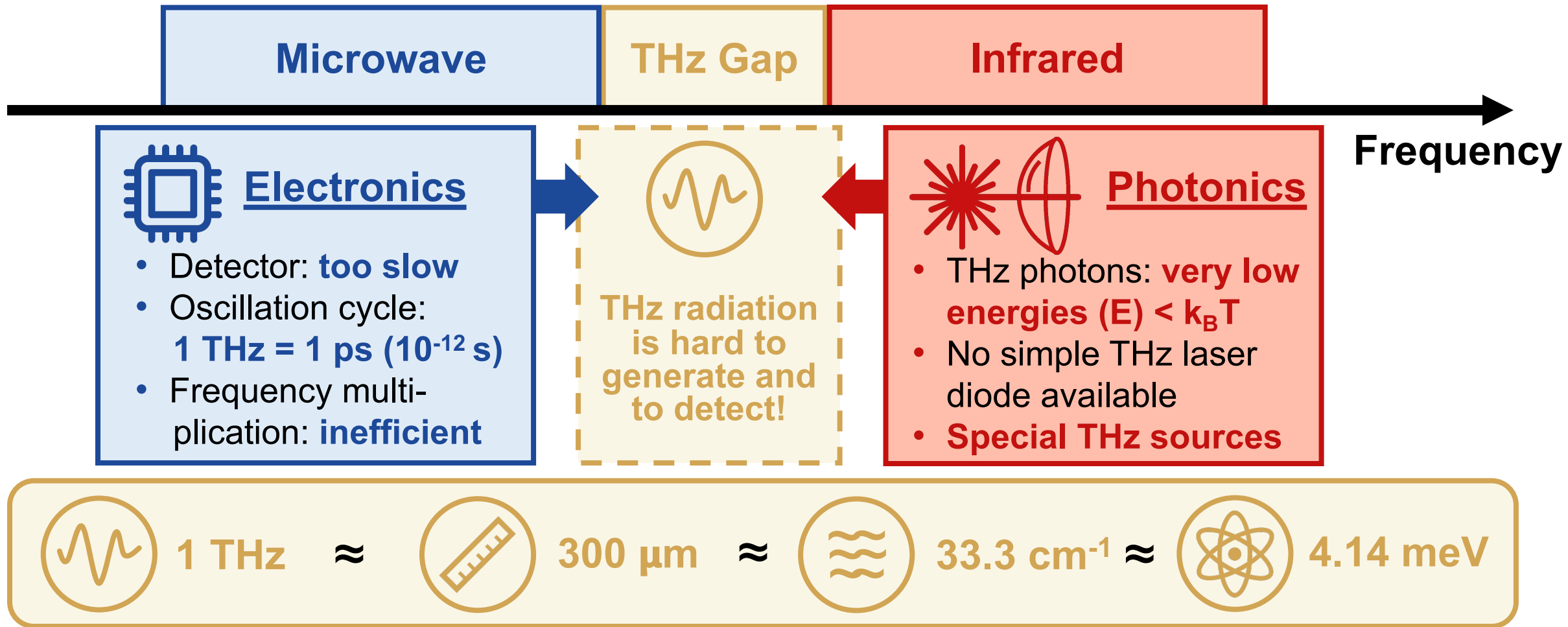


Locating the THz Region

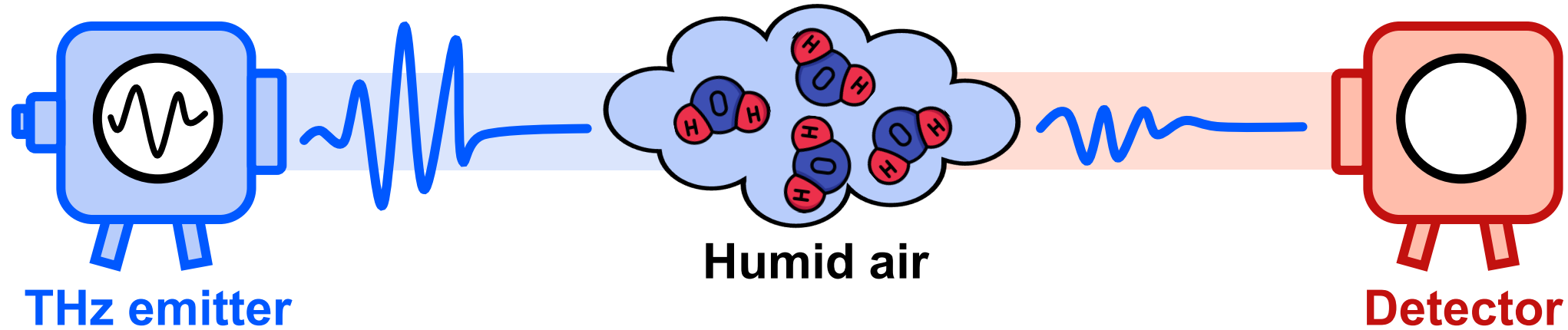


Why was the THz Region so Difficult to Access?

The THz region lies between two well-developed technology worlds.



Why was the THz Region so Difficult to Access?



Why H₂O vapour is a problem

- Frequency-selective absorption due to **rotational transitions** in the THz region
- **Weakens and distorts** the THz signal before it reaches the detector

Experimental challenge

- One does not only measure the sample, but **also the air**
- **Humidity changes** the measured THz spectrum

Practical solutions

- Use **short beam paths**
- Purge the beam path with **N₂**
- **Evacuate** the beam path

The THz Gap Explained

i.) Generating THz radiation is difficult:

It is hard to create strong and efficient THz sources with conventional microwave technologies.

ii.) Detecting THz radiation is difficult:

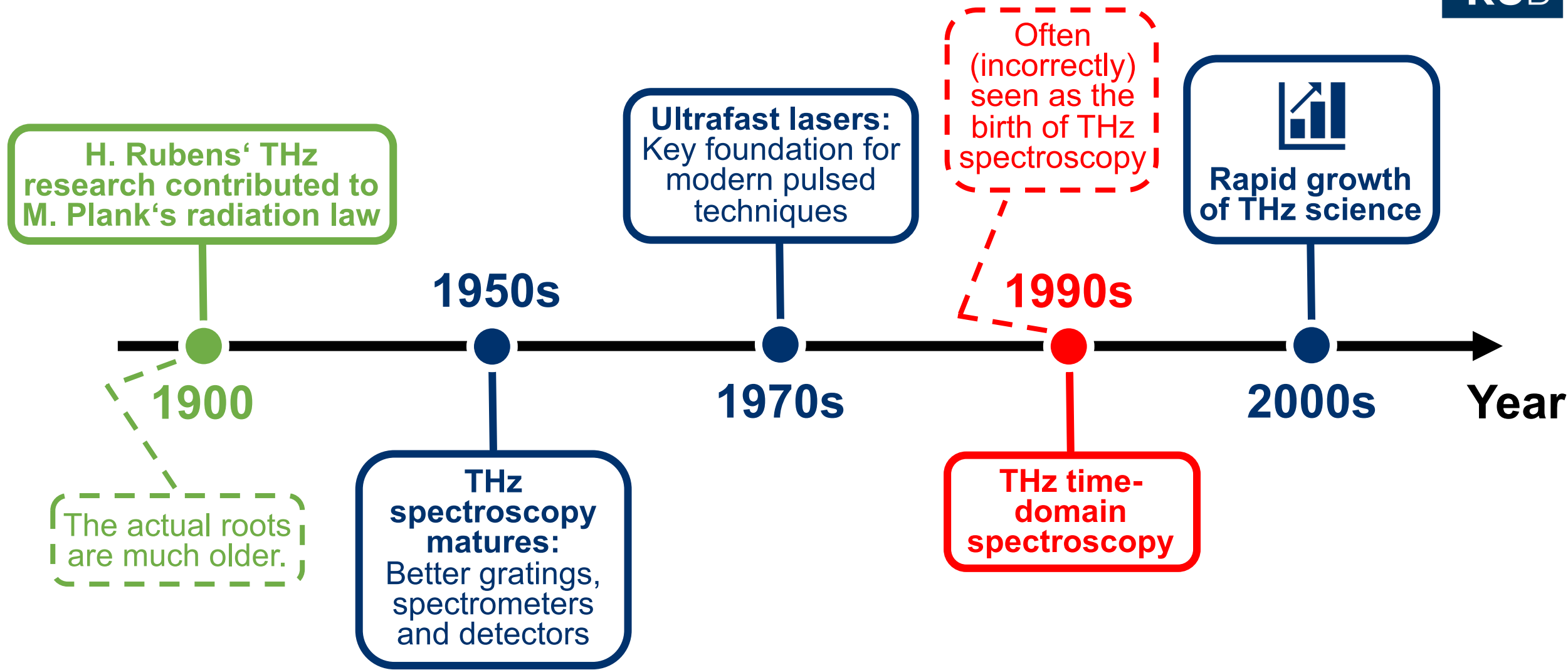
The low photon energy of THz radiation makes it challenging to measure with standard detectors.

iii.) THz radiation is strongly absorbed by H₂O vapor:

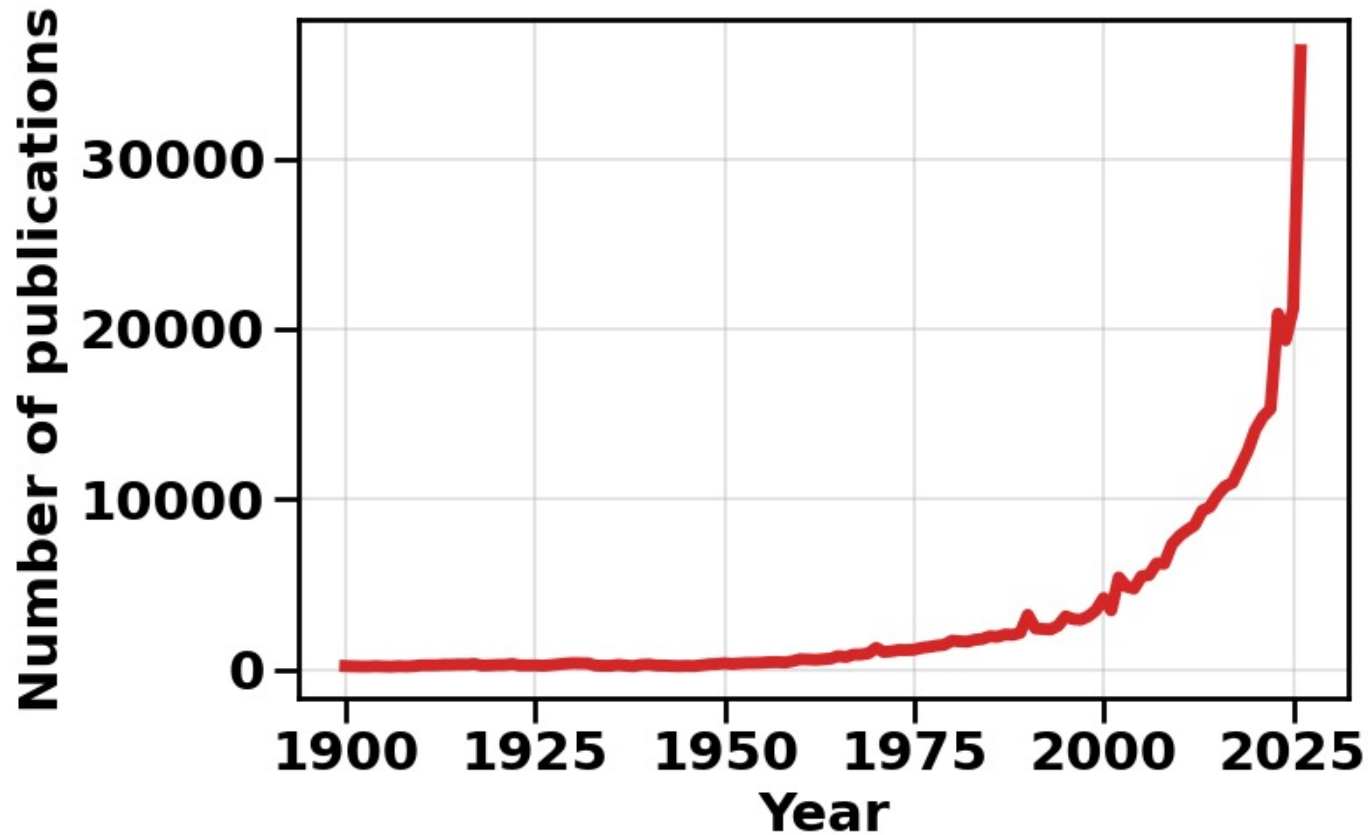
Even normal laboratory air will become part of the experiment by absorbing parts of the THz beam.

→ **How (and when) did the gap become accessible?**

History of THz Spectroscopy



Growing Interest in Studying THz Spectroscopy



Rapid, almost **exponential growth** after the year 2000.

The field is still highly active today!

Data source: OpenAlex

Search query:

("terahertz spectroscopy" OR "THz spectroscopy" OR "terahertz time-domain spectroscopy" OR "THz-TDS" OR "far-infrared spectroscopy" OR "far infrared spectroscopy" OR „FIR“)

Data retrieved: 31 May 2026.

Motivation of THz Spectroscopy

Gas phase molecules



Question: Which rotational transitions identify a molecule?

Biomolecules



Question: How does the hydrogen-bond network change when a protein folds/unfolds?

Crystals & pharmaceuticals

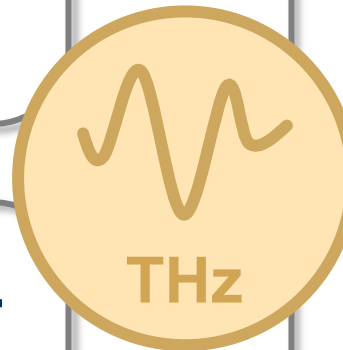


Question: Can THz spectroscopy help to distinguish different crystalline forms (polymorphs)?

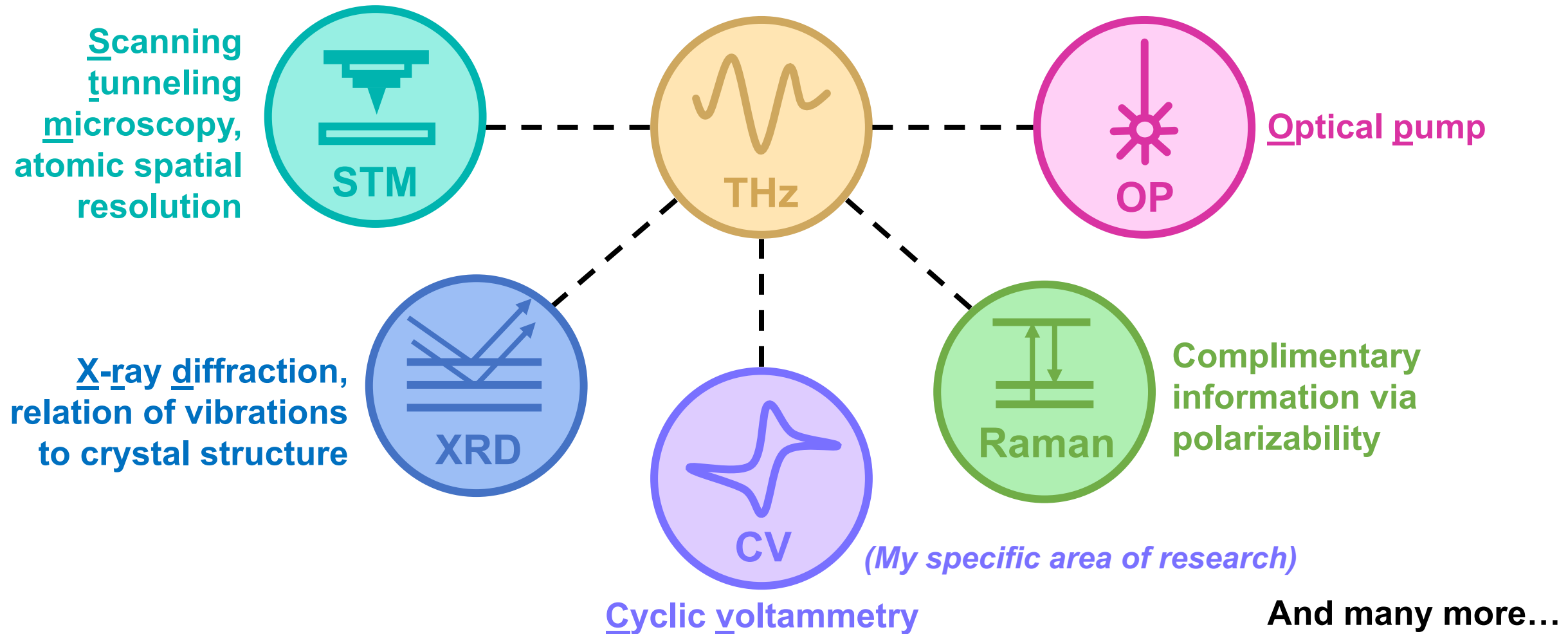
Functional materials



Question: How do charge carriers move after photoexcitation?



Combining THz Spectroscopy With Other Techniques



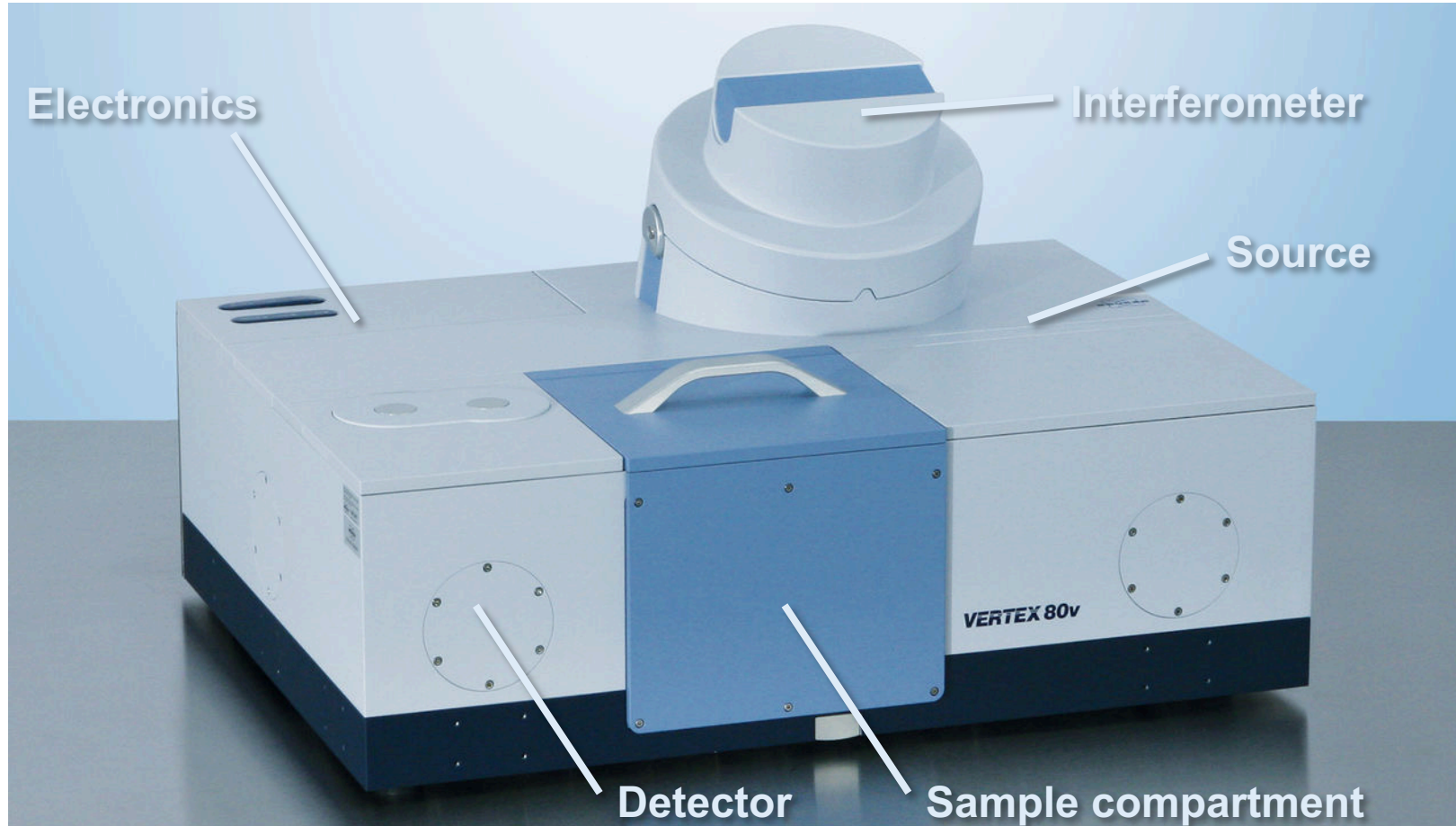
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Fourier-Transform Infrared (FTIR) Spectrometer

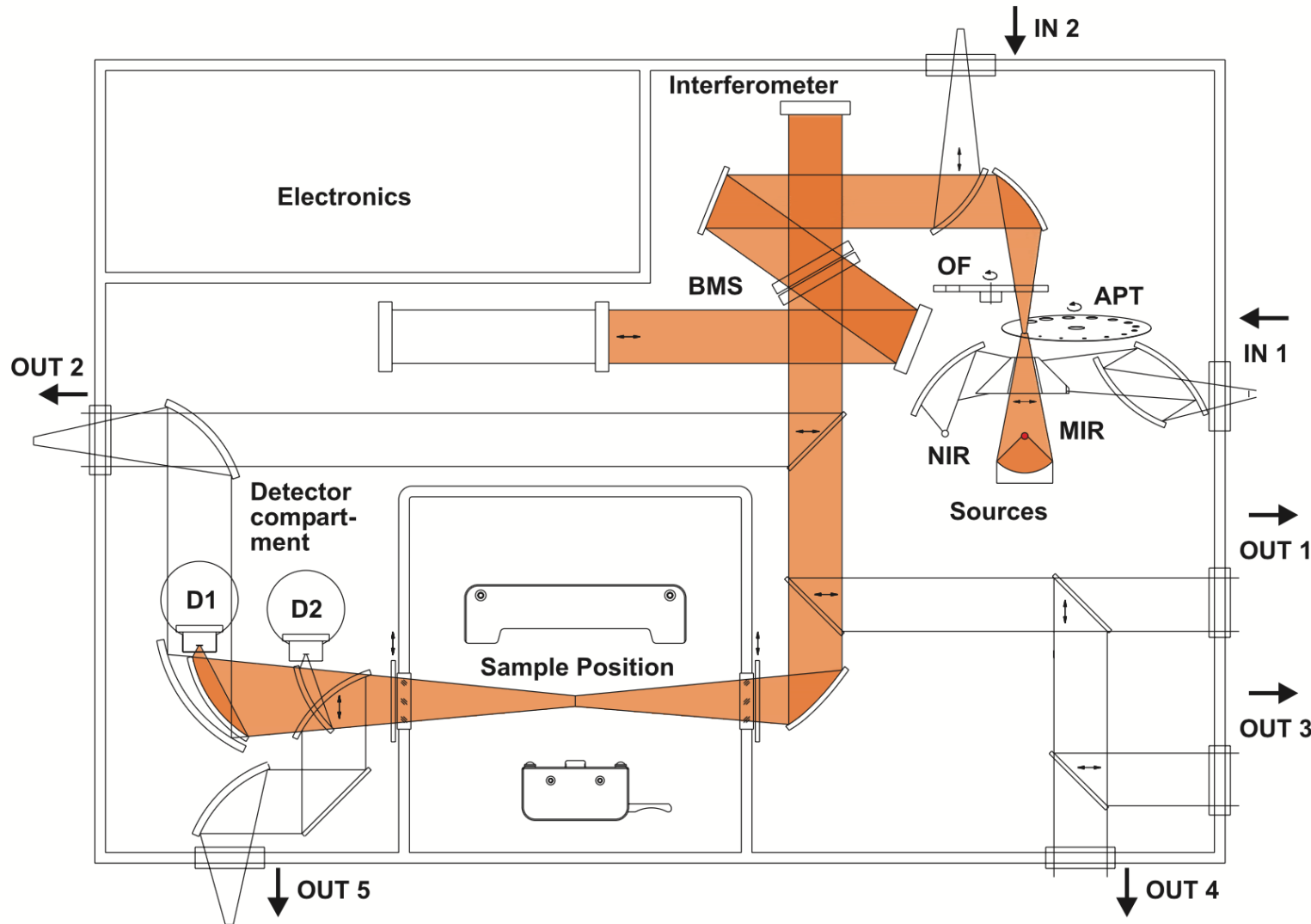


Key idea:
Detector records an **interferogram**.

The IR spectrum is then obtained mathematically by a **Fourier transform**.

Picture taken from: <https://www.bruker.com/de/products-and-solutions/infrared-and-raman/ft-ir-research-spectrometers/vertex-research-ft-ir-spectrometer/vertex-80-80v-ft-ir-spectrometer.html>

A Look Inside...

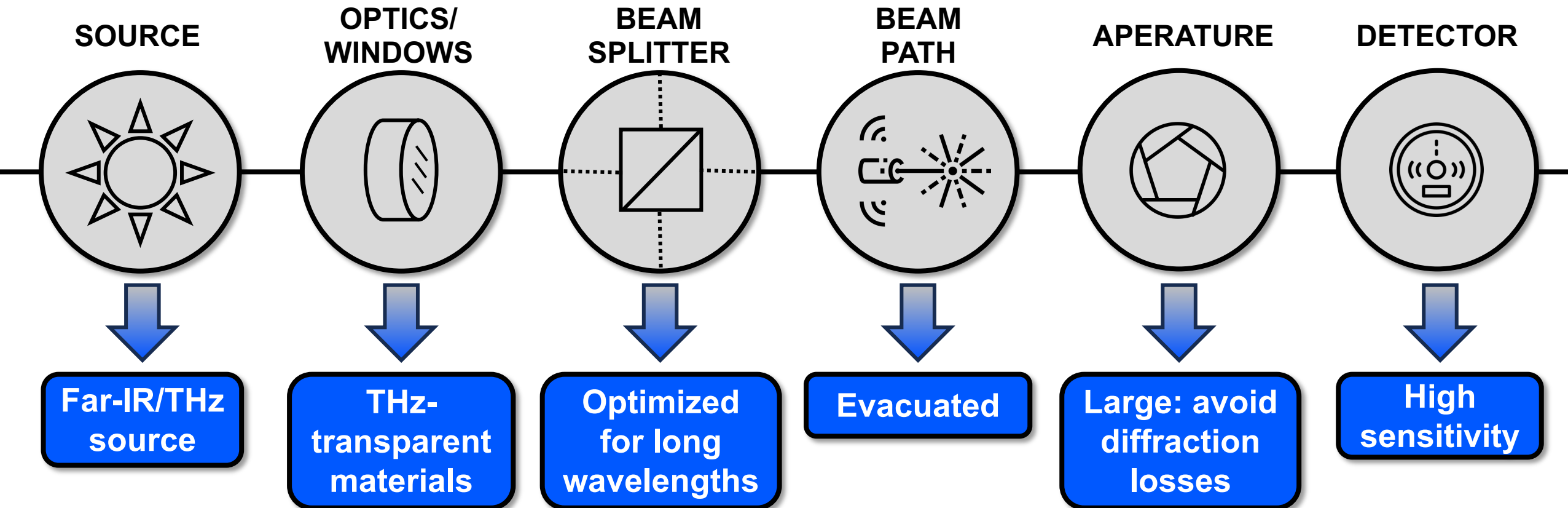


BMS: beam splitter
APT: aperture
OF: optical filter
IN 1, 2: external beam ports
OUT 1, 3, 4: beam exit ports
OUT 2, 5: external detectors
D1/D2: detectors

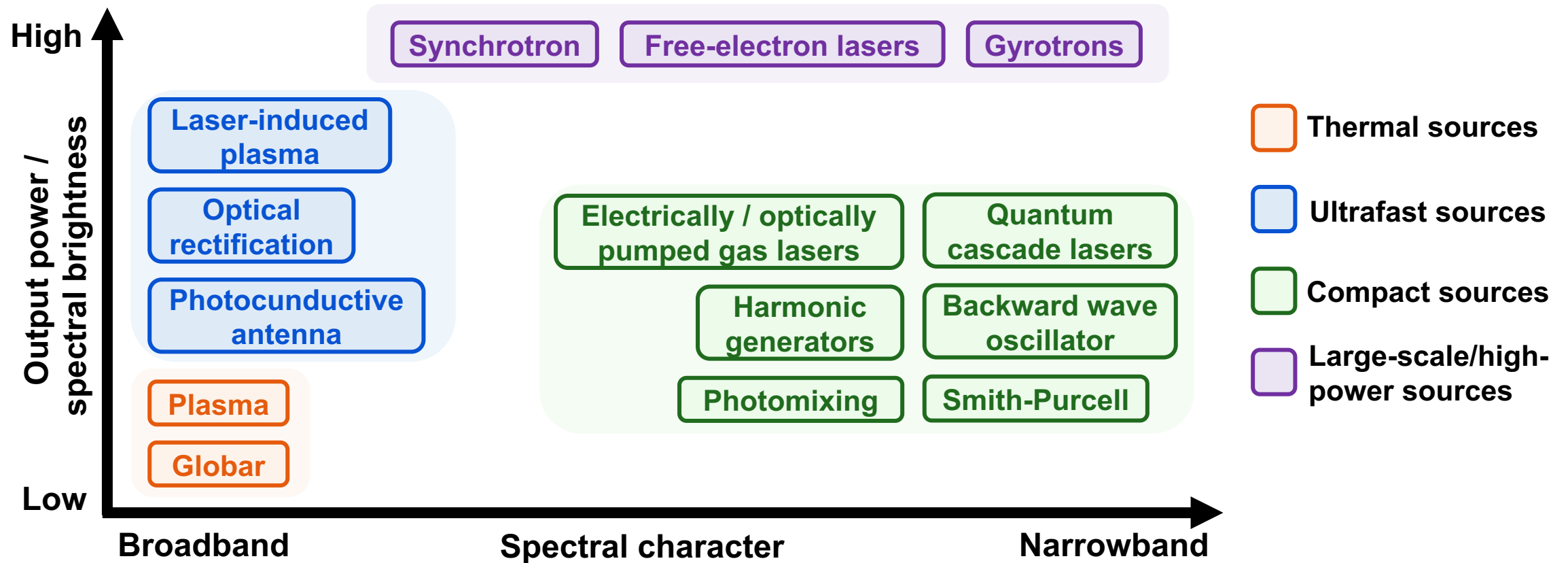
Picture taken from: VERTEX80_Flyer_EN.pdf (Bruker Homepage about Vertex 80/80v)

How Can we Measure a THz Spectrum?

Difference to a “standard” FTIR spectrometer:
Hardware must be adapted to the THz range!



THz Source Landscape



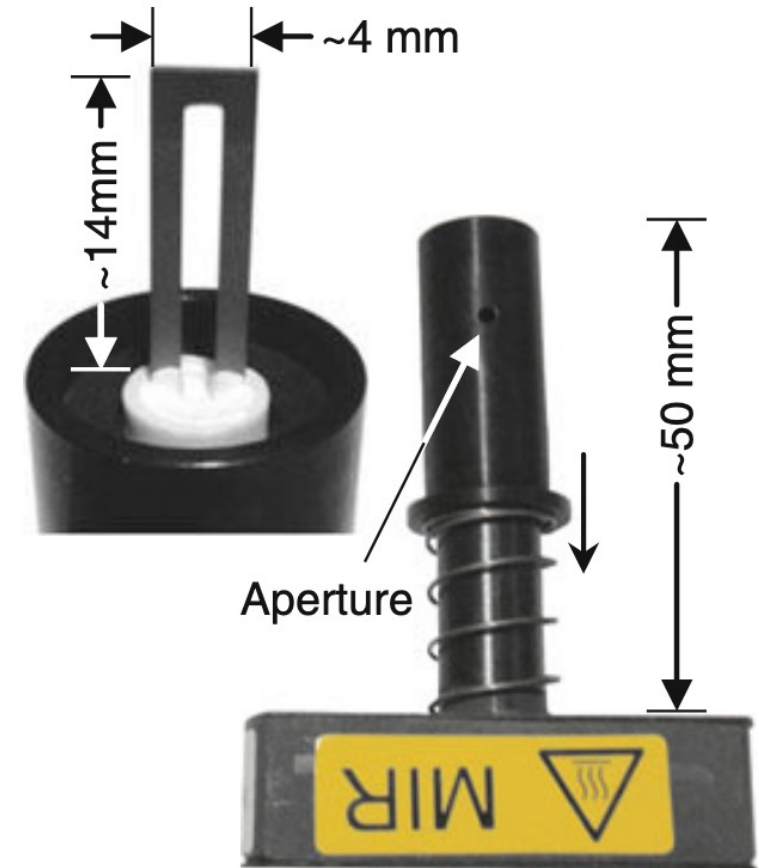
No single THz source is best for everything!
→ Choice depends on power, coherence, tunability and experimental scale.

Generation of THz Radiation

1.) Thermal sources:

a.) Globar

- Cylindrical rod of **SiC**
 - Capped with metallic electrodes
 - Needs a **water-cooled housing**
 - Operating temperatures: **1350-1500 K**
- Emits **broadband** thermal radiation
- Limitation: weak, incoherent THz radiation



Generation of THz Radiation

1.) Thermal sources:

b.) Plasma

- Electrical discharge in Hg vapor
- Hot plasma **emits broadband far-IR/THz radiation**
- **More powerful radiation** than the Globar
 - ~ 8 x more powerful* from $5 - 30 \text{ cm}^{-1}$
 - $\sim 2\text{-}3$ x more powerful* from $30 - 100 \text{ cm}^{-1}$
- Limitation: still incoherent THz radiation

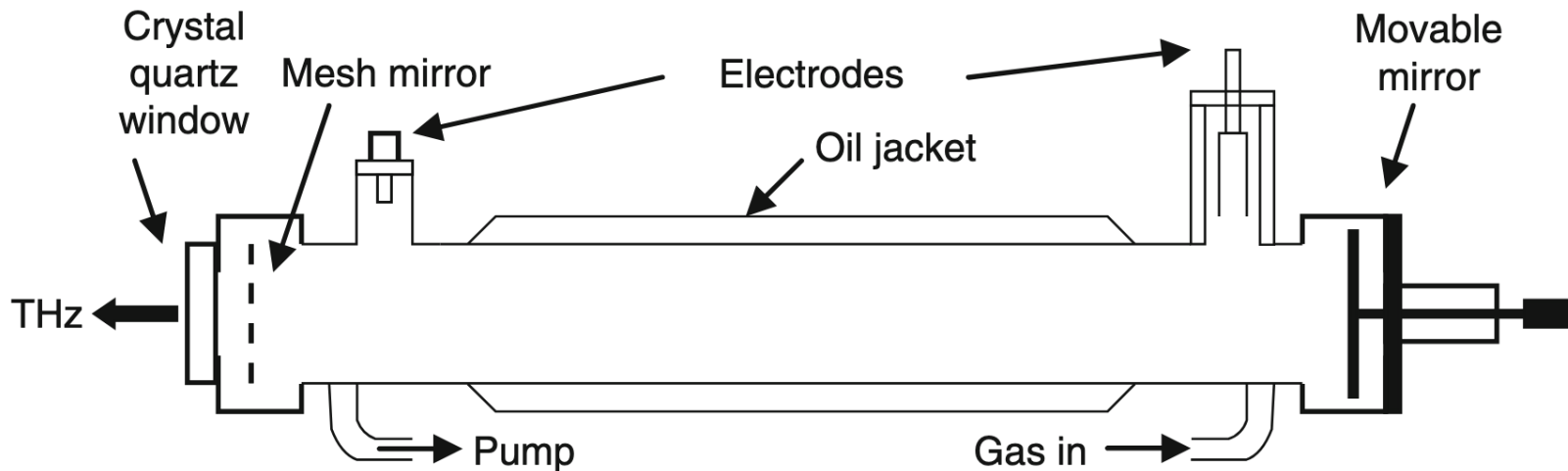


* Shumyatsky, P., Alfano, R. R., Terahertz sources. *Journal of Biomedical Optics* 2011, 16(3), 033001.

2.) Gas lasers:

a.) Electrically excited

- Generates THz radiation using a **gas discharge** inside a laser tube
- Low-pressure gas is placed inside a tube between two electrodes
- Apply **high voltage** → electrons are accelerated and collide with gas molecules and transfer energy → **energy population inversion**

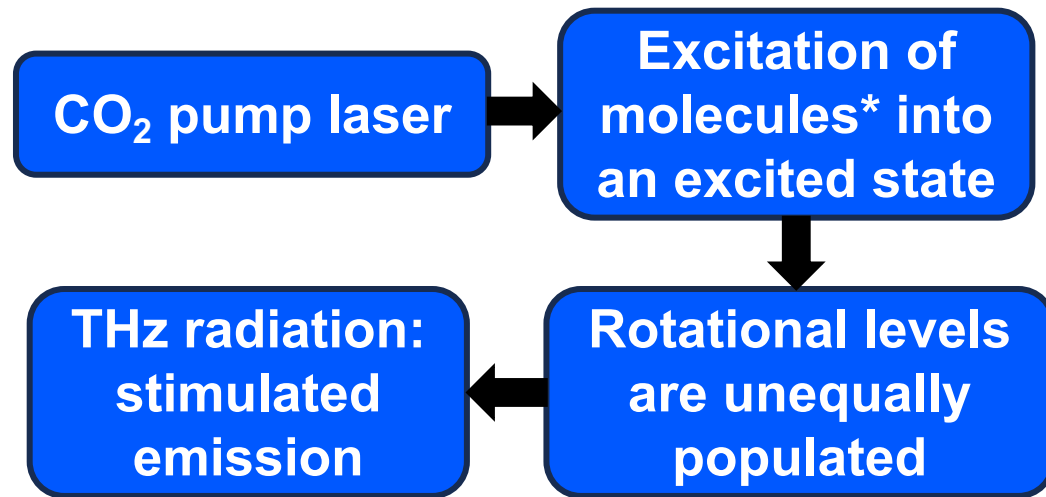


**Stimulated emission
between molecular
energy levels
produces coherent
THz radiation.**

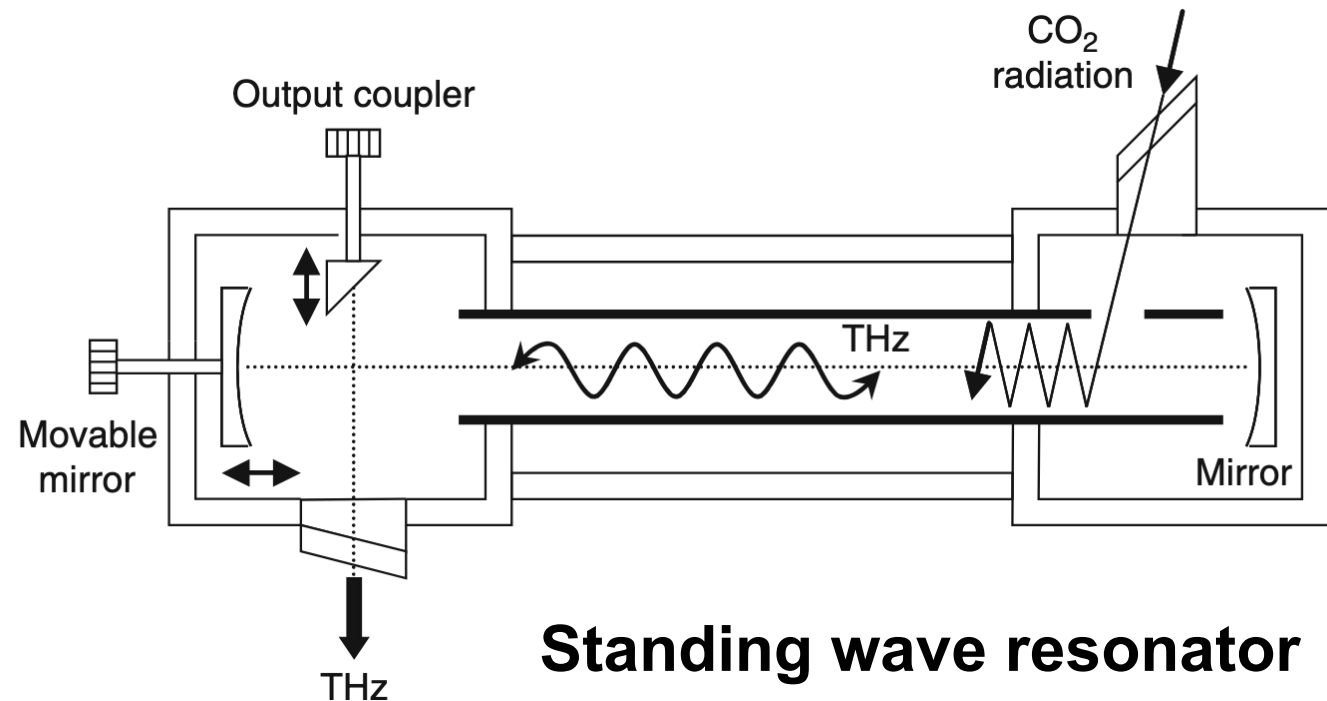
2.) Gas lasers:

b.) Optically excited

- Introduced in 1970, but **still used in many labs** nowadays
- Cover the frequency range from **0.15 – 8 THz**, output: up to **100 mW**

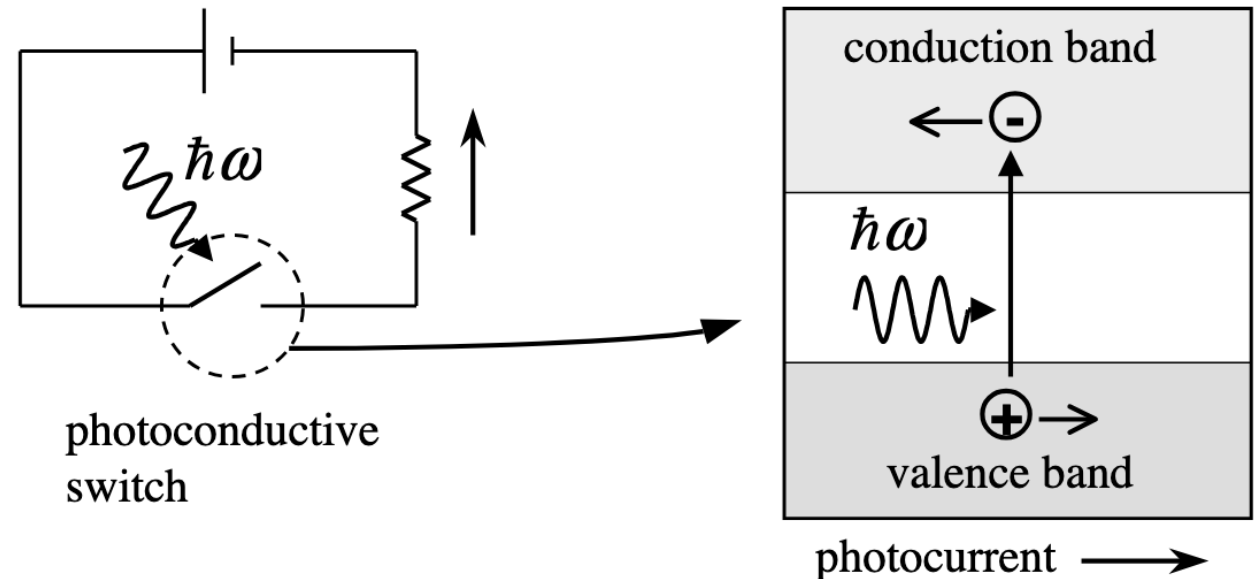


* Molecules are required to have a large, permanent dipole moment (e.g., CH₃OH, CH₂F₂, HCOOH, ...).



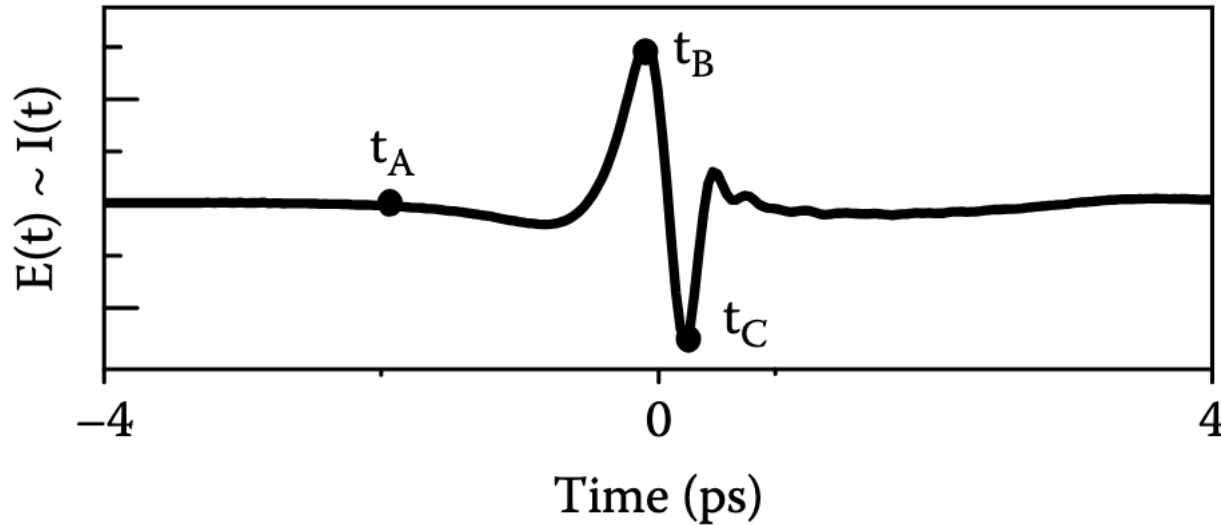
3.) Photoconductive antenna:

- A photoconductive antenna is an **electrical switch** exploiting the increase in **electrical conductivity** of semiconductors and insulators when they are exposed to **light**.
- Photoconductivity: increase in the number of free carriers
- $E_{\text{photon}} > E_{\text{bandgap}}$
- Switching action: **< 1 ps**
- Materials:
 - low-temperature grown gallium arsenide (LT-GaAs),
 - radiation-damaged silicon-on-sapphire (RD-SOS)



Detection of THz Radiation

1.) Photoconductive antenna:



t : delay time between gate and THz pulse

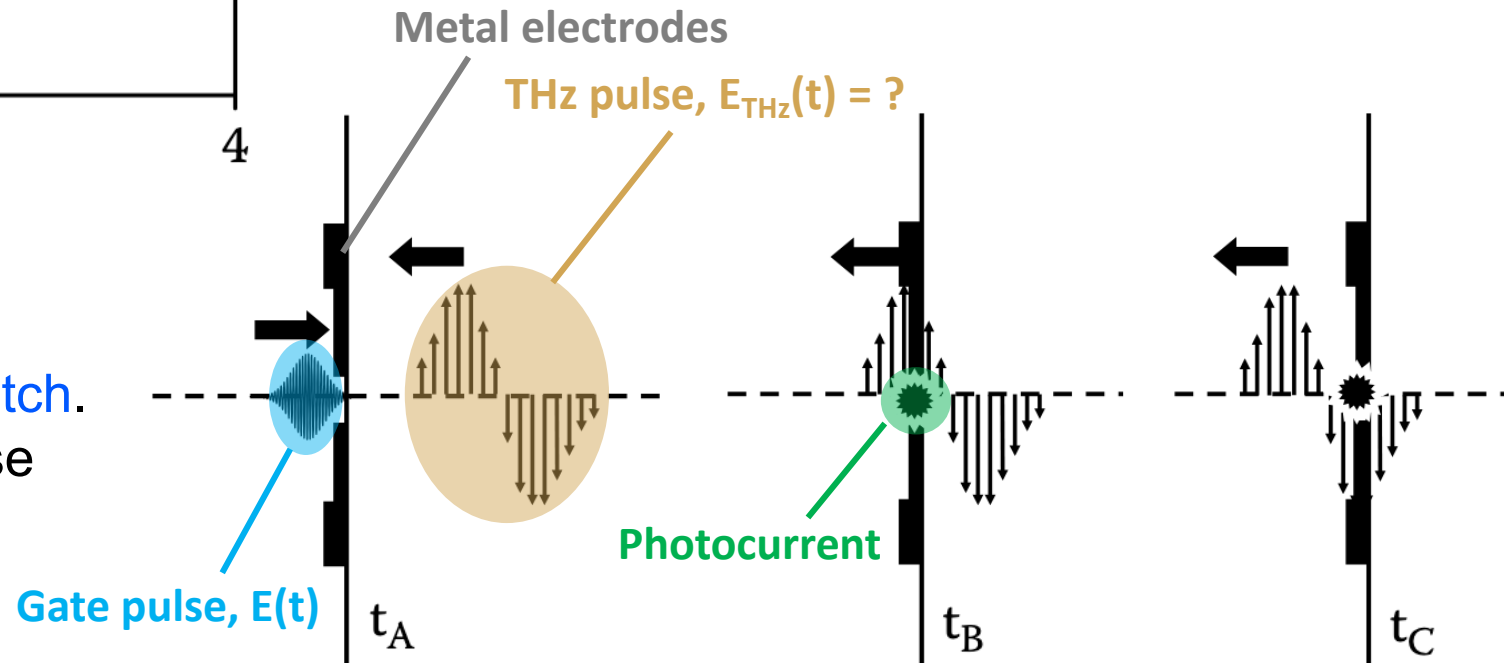
t_A : gate pulse far away $\rightarrow$ no signal

t_B : samples positive THz field $\rightarrow$ positive current

t_C : samples negative THz field $\rightarrow$ negative current

Key idea:

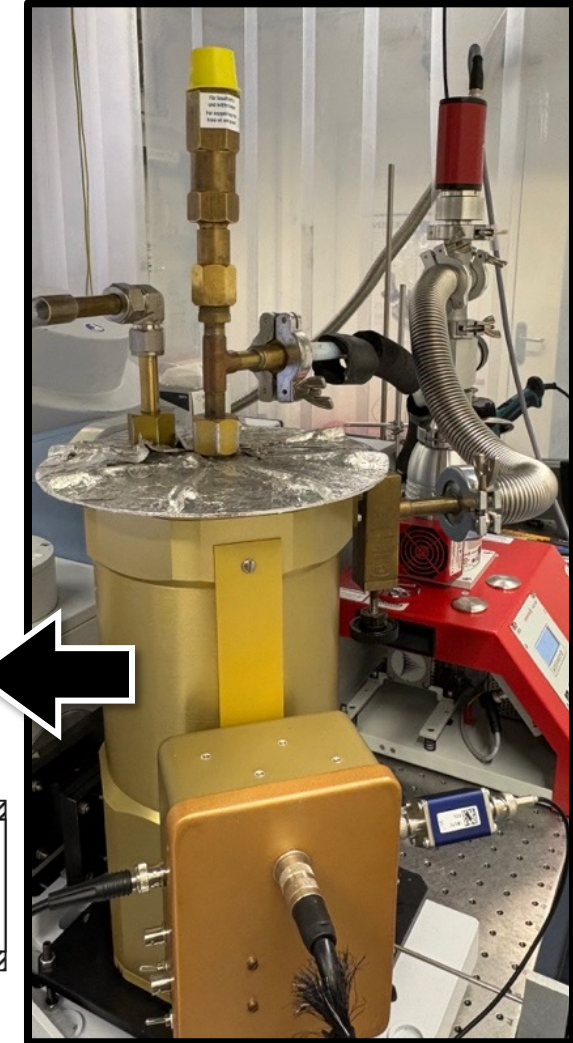
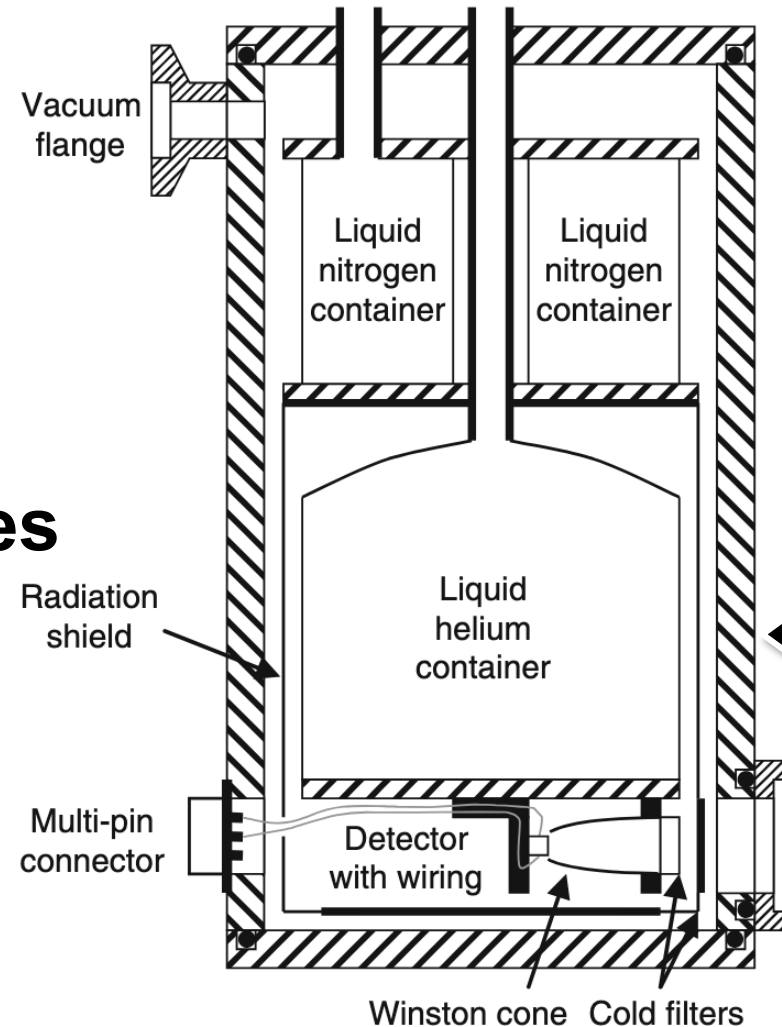
The fs gate pulse acts as an **ultrafast switch**. By **changing the delay** between THz pulse and gate pulse, the **full THz waveform**, $E_{\text{THz}}(t)$, can be reconstructed.



Detection of THz Radiation

2.) Bolometer: a **thermal detector** that measures **electromagnetic radiation** by absorbing it and detecting the resulting **temperature increase**

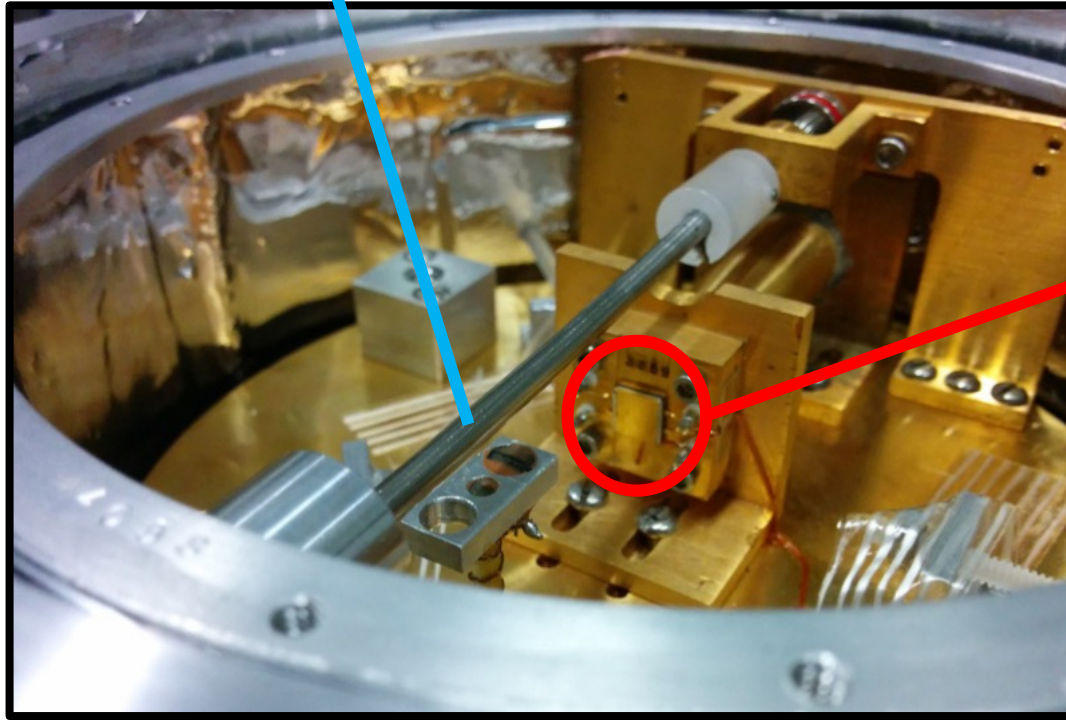
- Temperature change modifies **electrical resistance** of the detector element
- Operated at **cryogenic temperatures** ($T < 4.2 \text{ K}$)
[He(I): low thermal noise]



Detection of THz Radiation

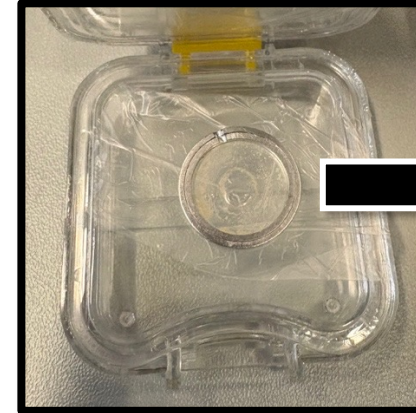
Opened detector housing of our liquid-He-cooled Si bolometer:

Control shaft: switching filter on/off

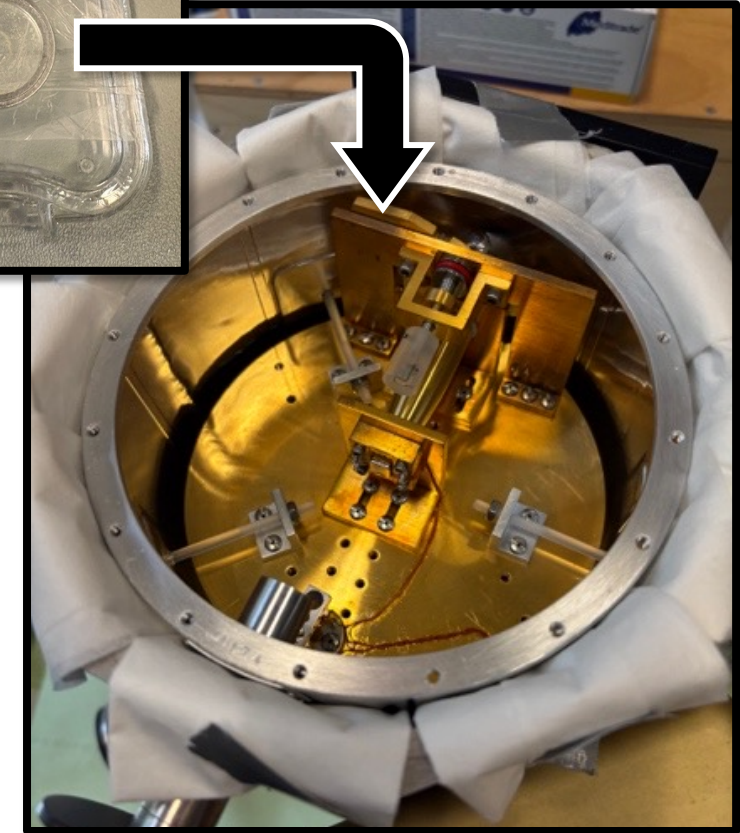


Diamond absorber: collects THz radiation and heats up slightly

A broken PE filter ☹



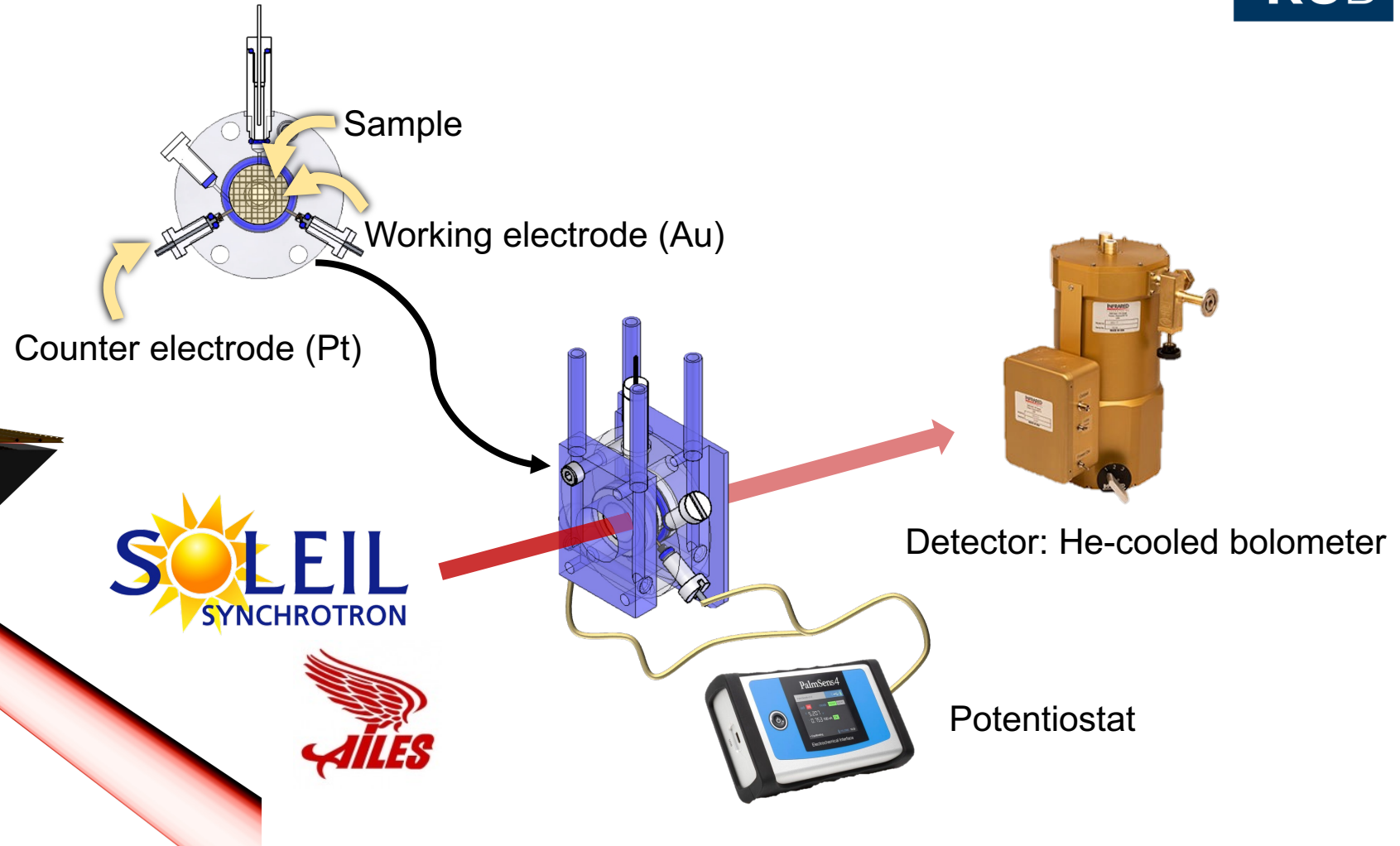
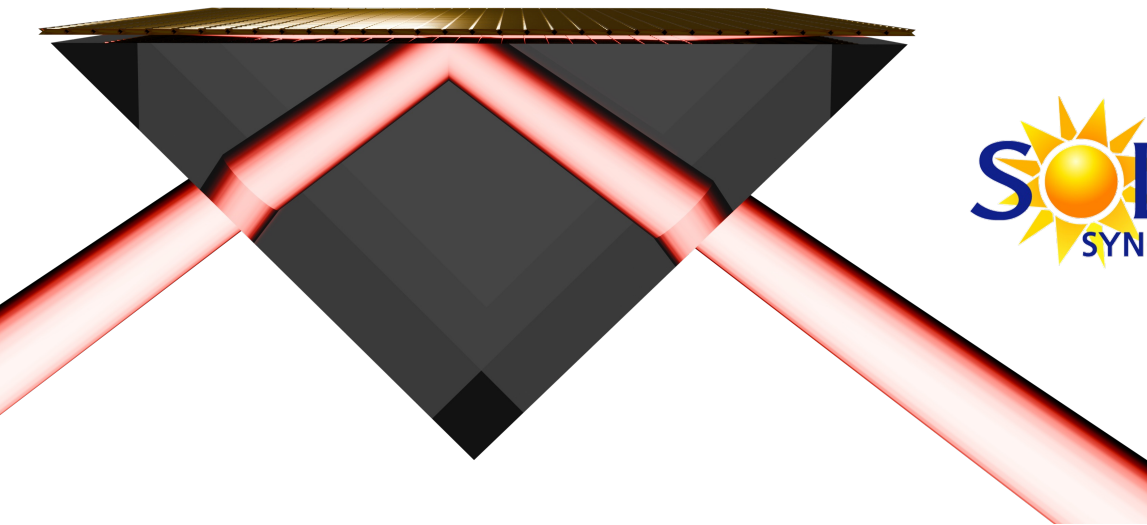
Blocks unwanted VIS and NIR light



Example 1 – Spectroelectrochemistry

Problem: conventional metal electrodes would block the IR radiation

Solution: gold mesh



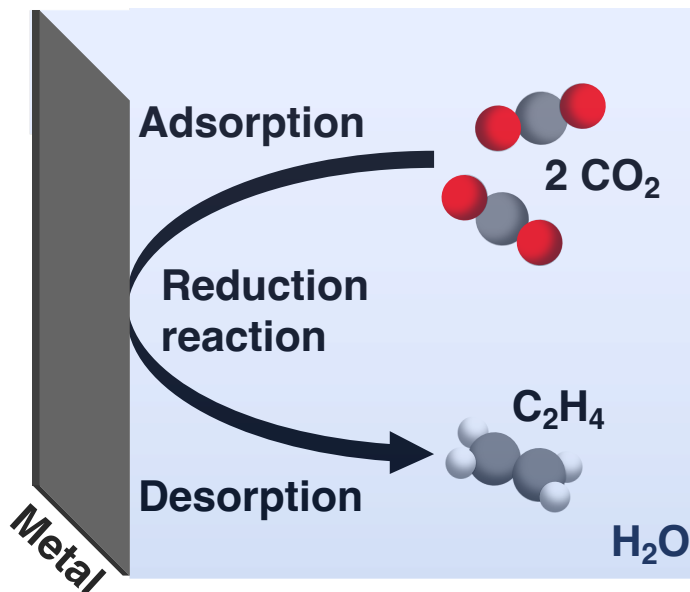
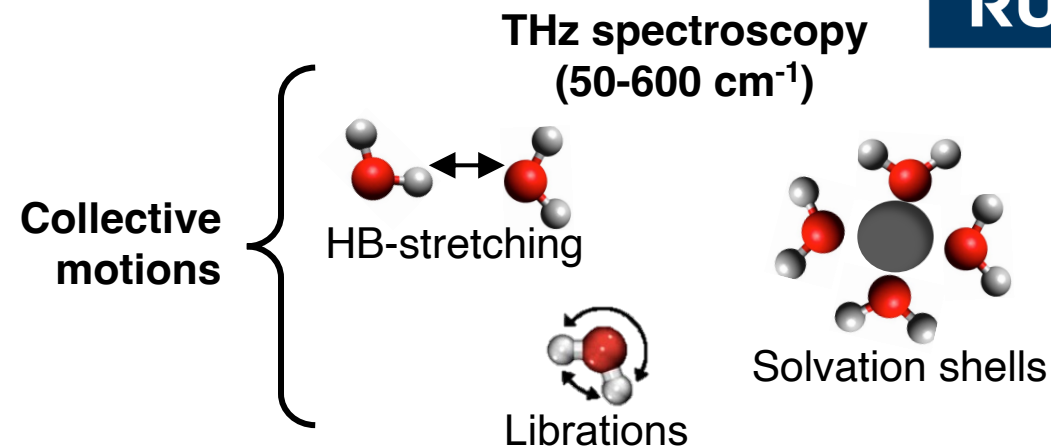
S.R. Alfarano, S. Pezzotti, C.J. Stein, Z. Lin, F. Sebastiani, S. Funke, C. Hoberg, I. Kolling, C.Y. Ma, K. Mauelshagen, T. Ockelmann, G. Schwaab, L. Fu, J.-B. Brubach, P. Roy, M. Head-Gordon, K. Tschulik, M.-P. Gaigeot, M. Havenith, *PNAS* **118**, e2108568118 (2021).

Example 1 – Spectroelectrochemistry

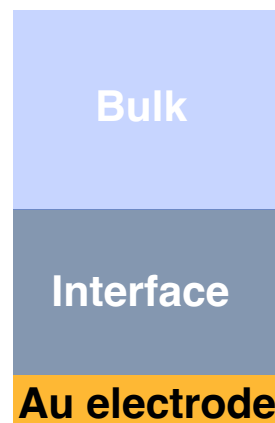
What can be observed with such a setup?

THz spectroscopy: ideal to probe solvation shells, but it is a bulk-sensitive technique!

- Processes governing electrochemical reactions
- Solvation & interfacial water dynamics



THz spectrum at V_1



THz spectrum at V_2



Difference THz spectrum: $V_1 - V_2$

=

Interface — Interface

Bulk contribution cancels out!

S.R. Alfarano, S. Pezzotti, C.J. Stein, Z. Lin, F. Sebastiani, S. Funke, C. Hoberg, I. Kolling, C.Y. Ma, K. Mauelshagen, T. Ockelmann, G. Schwaab, L. Fu, J.-B. Brubach, P. Roy, M. Head-Gordon, K. Tschulik, M.-P. Gaigeot, M. Havenith, *PNAS* **118**, e2108568118 (2021).

Example 1 – Spectroelectrochemistry

THz fingerprints: rattling modes

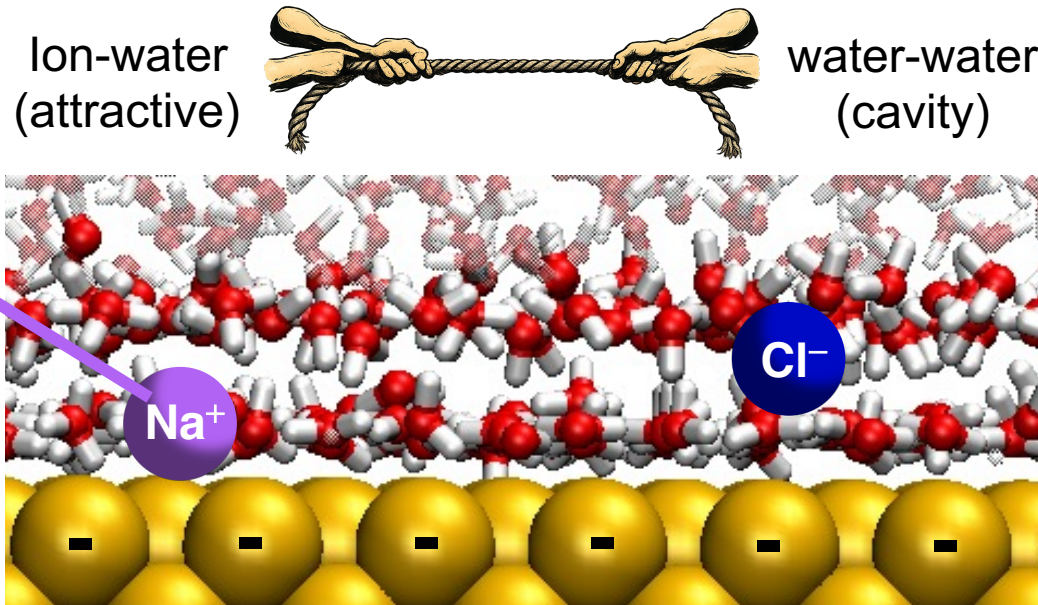
→ Overlapping contributions from ions & H₂O

Solution: Principal component analysis (PCA)

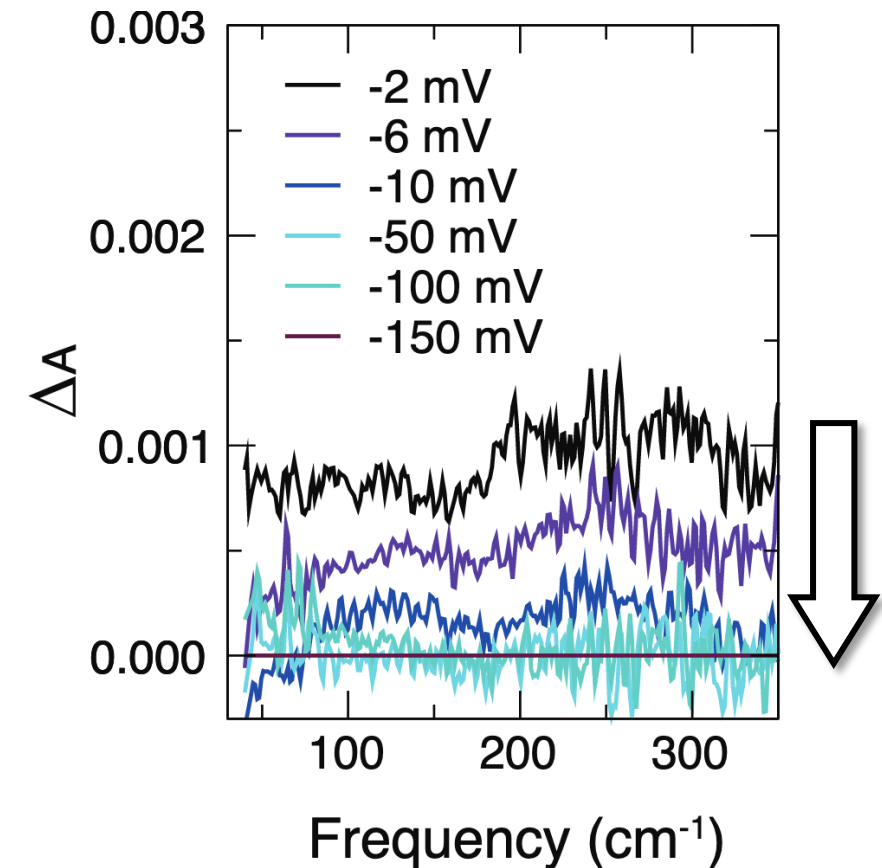
Looses its
hydration
shell quickly

Hydrophobic {

Hydrophilic {



Negative (cathodic) potential sweep

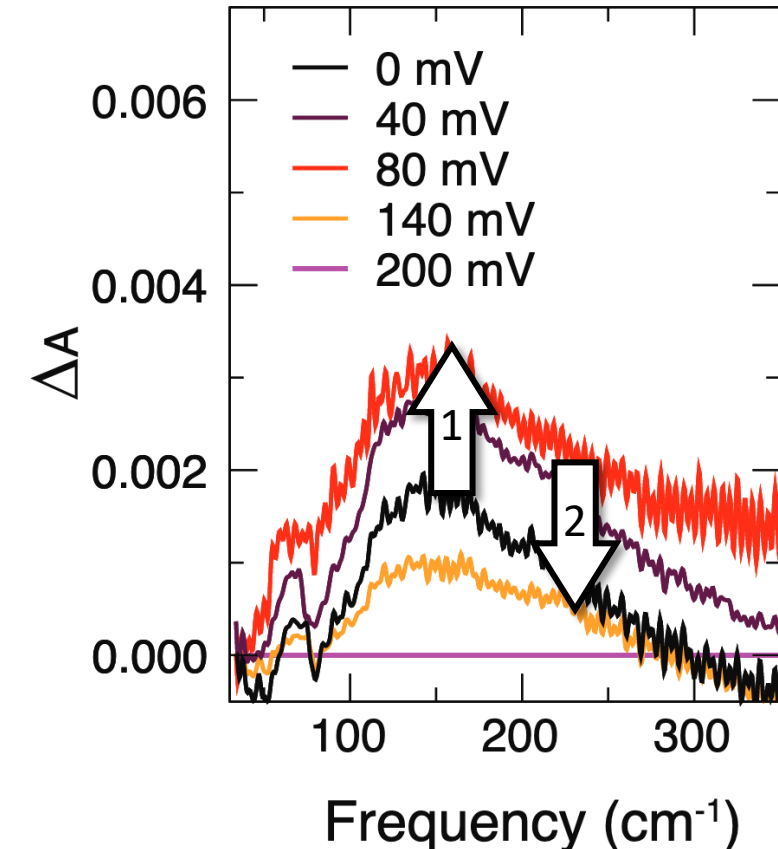
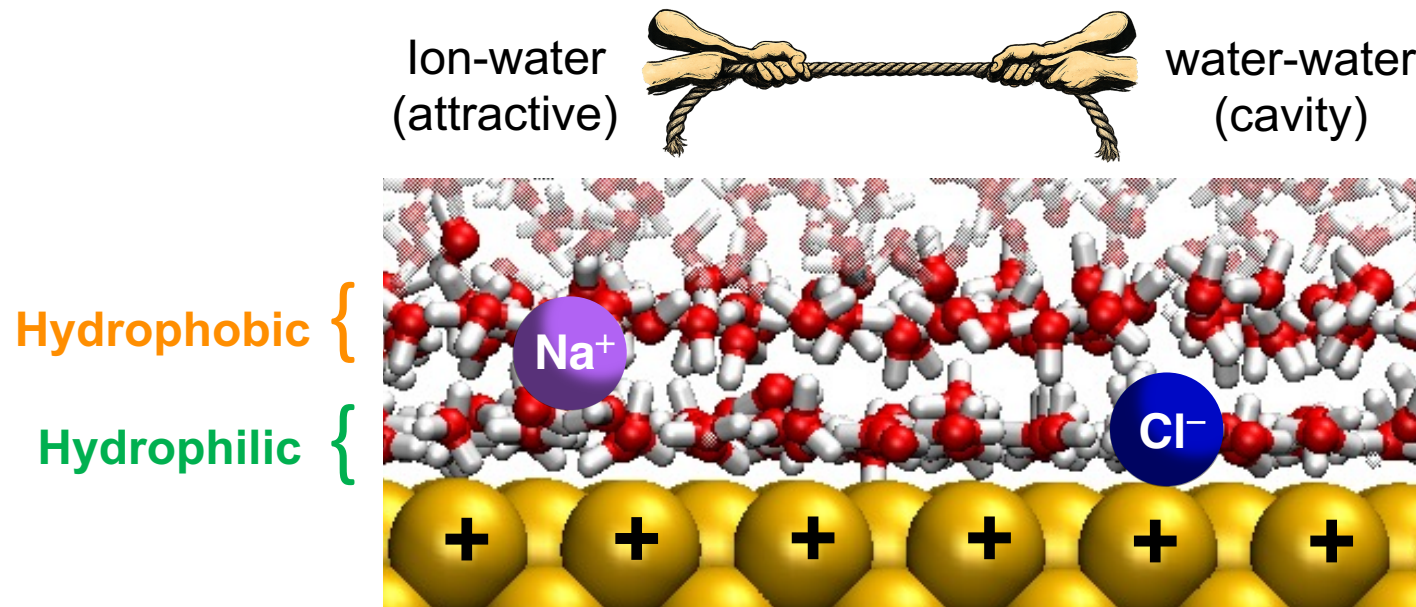


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Example 1 – Spectroelectrochemistry

- Cl^- remains hydrated over a larger potential window
- Hydration shell is only removed at higher potentials ($> 80 \text{ mV}$)

Positive (anodic) potential sweep



S.R. Alfarano, S. Pezzotti, C.J. Stein, Z. Lin, F. Sebastiani, S. Funke, C. Hoberg, I. Kolling, C.Y. Ma, K. Mauelshagen, T. Ockelmann, G. Schwaab, L. Fu, J.-B. Brubach, P. Roy, M. Head-Gordon, K. Tschulik, M.-P. Gaigeot, M. Havenith, *PNAS* **118**, e2108568118 (2021).

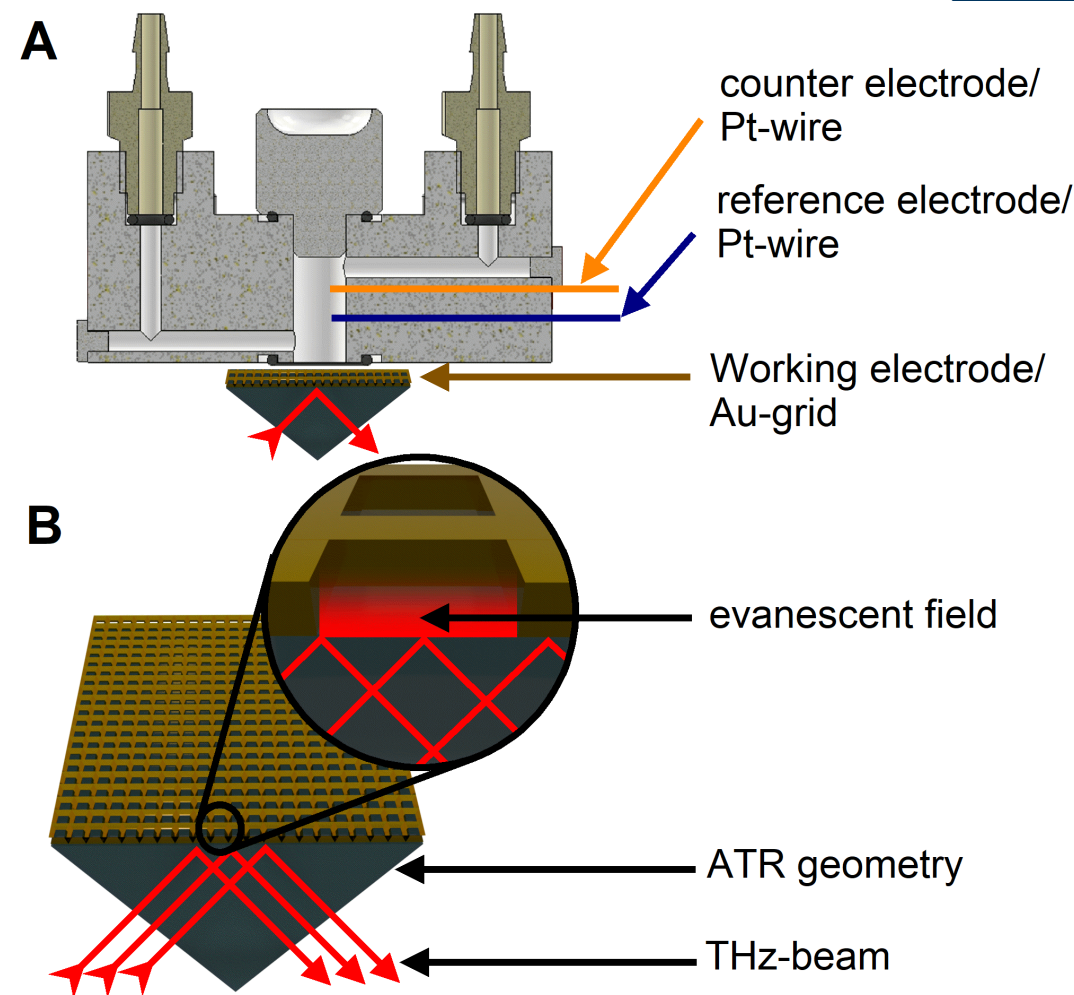
Example 2 – Spectroelectrochemistry

THz-ATR spectroelectrochemistry

- 3 electrode setup
- Platinum wires as semi-reference and counter electrode
- Gold grid (working electrode) pressed on the diamond
- Currently operated under static electrolyte conditions

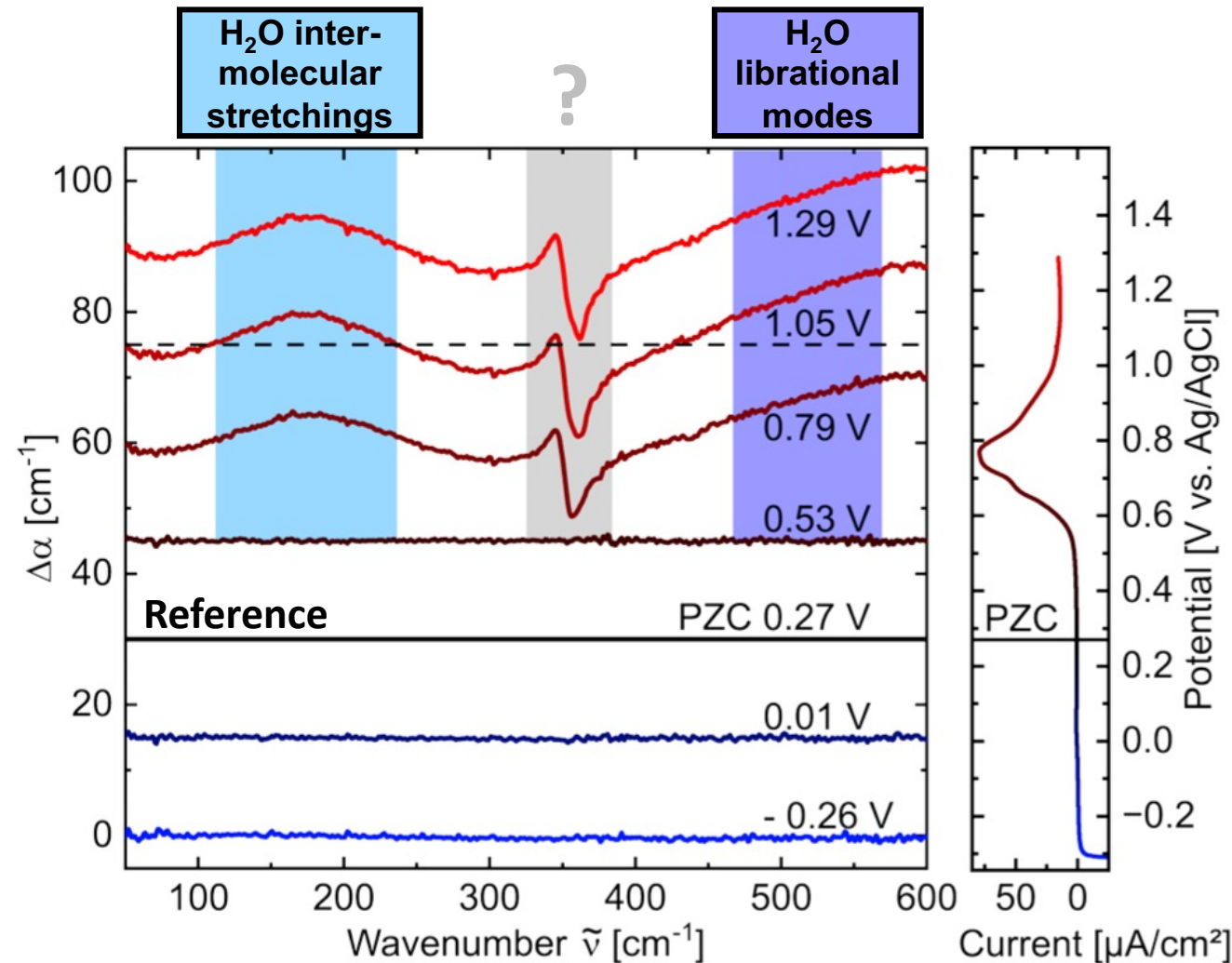
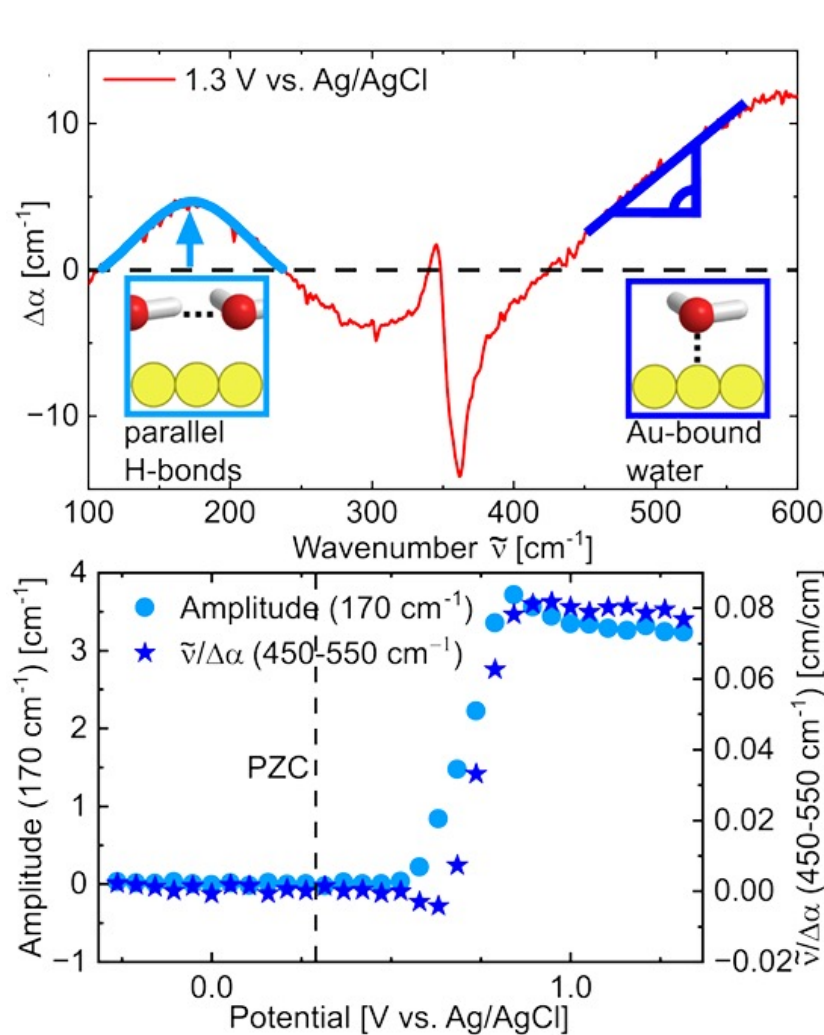
Tetrabutylammonium (TBA) cations

- Large hydrophobic ions → known to boost the hydrogen evolution reaction by a factor of 4
- Do they accumulate at the interface?
- Significant changes in H-bond network?



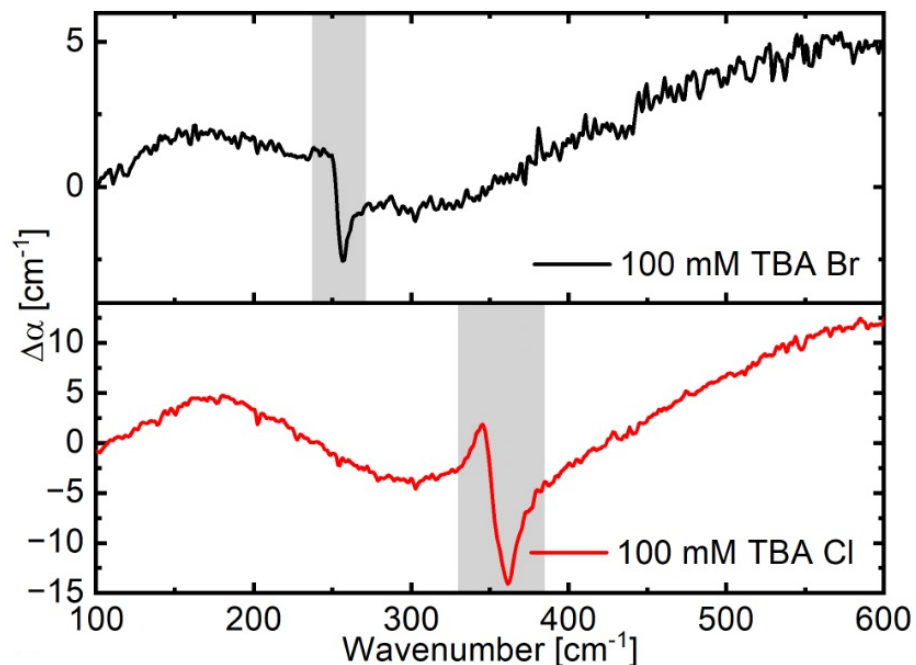
N.S. Wichmann, W. Chen, Q. Yao, X. Yu, M.B. Jassar, S. Pezzotti, M. Havenith, *JACS*, 148, 13, 13550-13560 (2026).

Example 2 – Formation of Thin Films

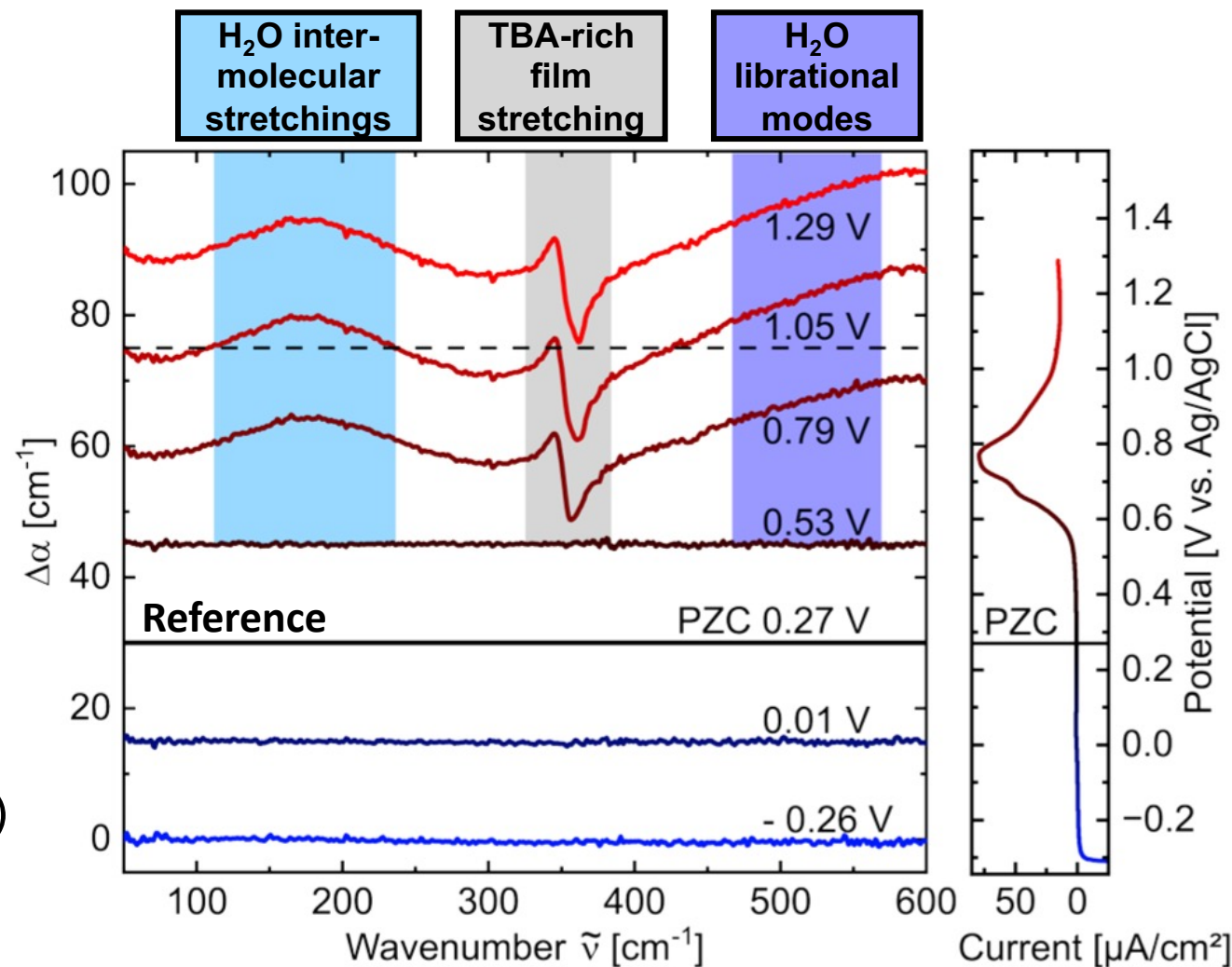


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Example 2 – Formation of Thin Films



- Additional measurement of TBA Br sample
- Band position depends on the anion (Cl^- vs Br^-)
- Indicates an ion-specific vibrational mode



N.S. Wichmann, W. Chen, Q. Yao, X. Yu, M.B. Jassar, S. Pezzotti, M. Havenith, *JACS*, 148, 13, 13550-13560 (2026).

Why Were Those Examples ‘Linear’?

We did not drive the system into a non-linear (excited) state because the **field strengths** of the THz pulses were **not high enough**.

The sample response is **proportional** to the incident THz field (weak field: a few V/cm to kV/cm).

Linear THz spectroscopy = weak-probe spectroscopy

But what happens if the THz field is no longer just a weak probe, but strong enough ($> \sim 10\text{-}100$ kV/cm) to actively drive the system?

3) Nonlinear THz Spectroscopy:

- 3.a.) From probing to driving: entering the nonlinear regime**
- 3.b.) Strong-field THz sources**
- 3.c.) Field-resolved THz detection**
- 3.d.) THz time-domain spectroscopy (THz-TDS)**
- 3.e.) Optical-pump-THz-probe spectroscopy and examples**

4.) Summary:

- 4.a.) Take-home messages
- 4.b.) Questions for self-studying
- 4.c.) Further reading

Entering the Nonlinear Regime

$$P(t) = \underbrace{\varepsilon_0 \left[\chi^{(1)} E(t) \right]}_{\text{Linear}} + \underbrace{\varepsilon_0 \left[\chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]}_{\text{Nonlinear}}$$

$P(t)$: induced polarization of the material

$E(t)$: applied THz electric field

ε_0 : vacuum permittivity

$\chi^{(1)}$: linear susceptibility

$\chi^{(2)}, \chi^{(3)}$: nonlinear susceptibilities

t : time

From probing to driving:

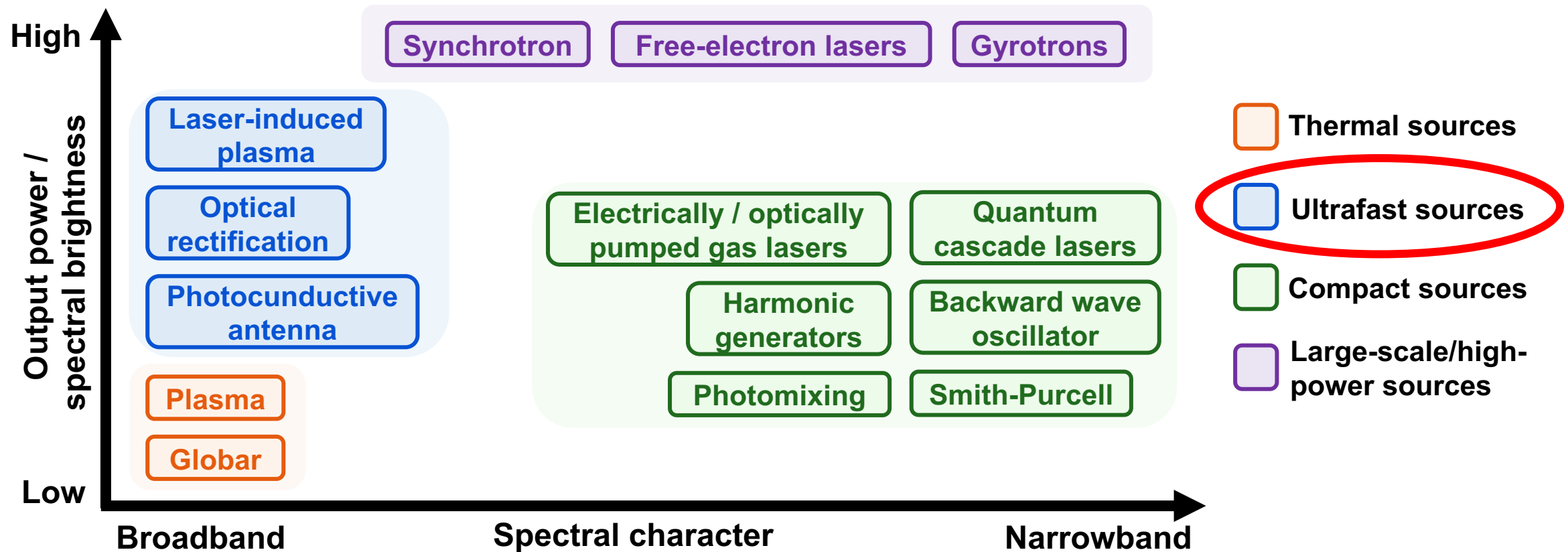
Weak THz field → measures equilibrium response

Strong THz field → modifies the material response (> 2 photons involved)

Linear regime: the material response is proportional to the electric field

Nonlinear regime: higher-order terms contribute to the polarization

THz Source Landscape (Nonlinear)



In the lab, nonlinear THz experiments are based on ultrafast laser-driven sources (high power)!
→ Short THz pulses are naturally broadband and provide high peak fields for field-resolved dynamics.

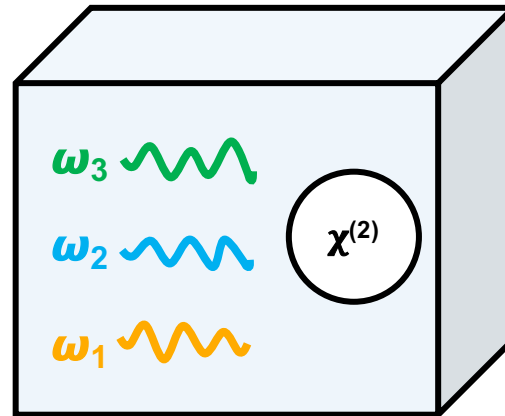
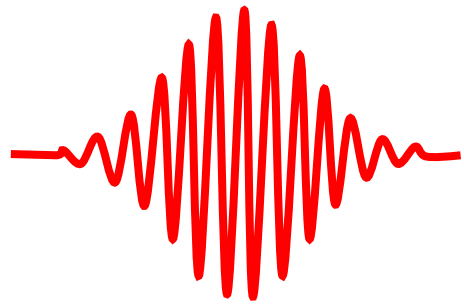
THz Source: Optical Rectification

- Optical rectification uses a **non-linear material** to generate THz radiation.

inorganic crystals (LiNbO₃, ZnTe, GaAs) or organic crystals

- An **ultrashort optical laser pulse** induces a **time-dependent polarization** in the crystal, which converts a photon into two photons.

Ultrashort optical pulse



Crystal



THz pulse



Lower frequency
(longer wavelength)

Frequency mixing:

$$\omega_{\text{THz}} = \omega_1 - \omega_2$$



IR pulse

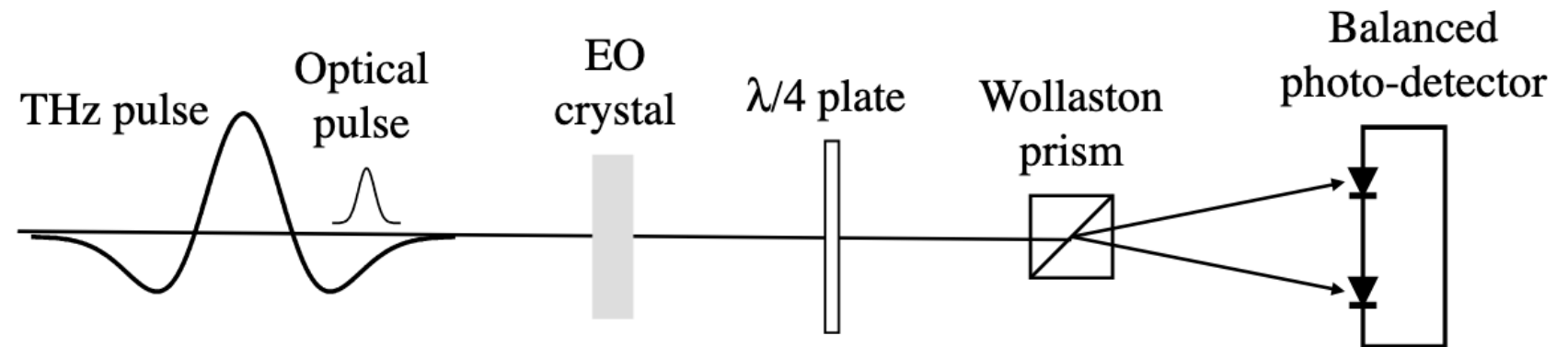
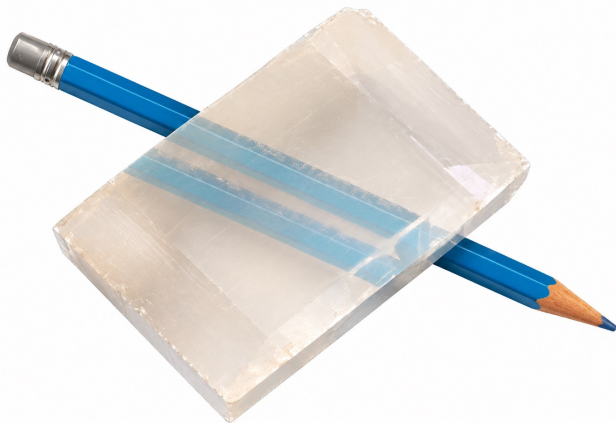


ω_{IR}

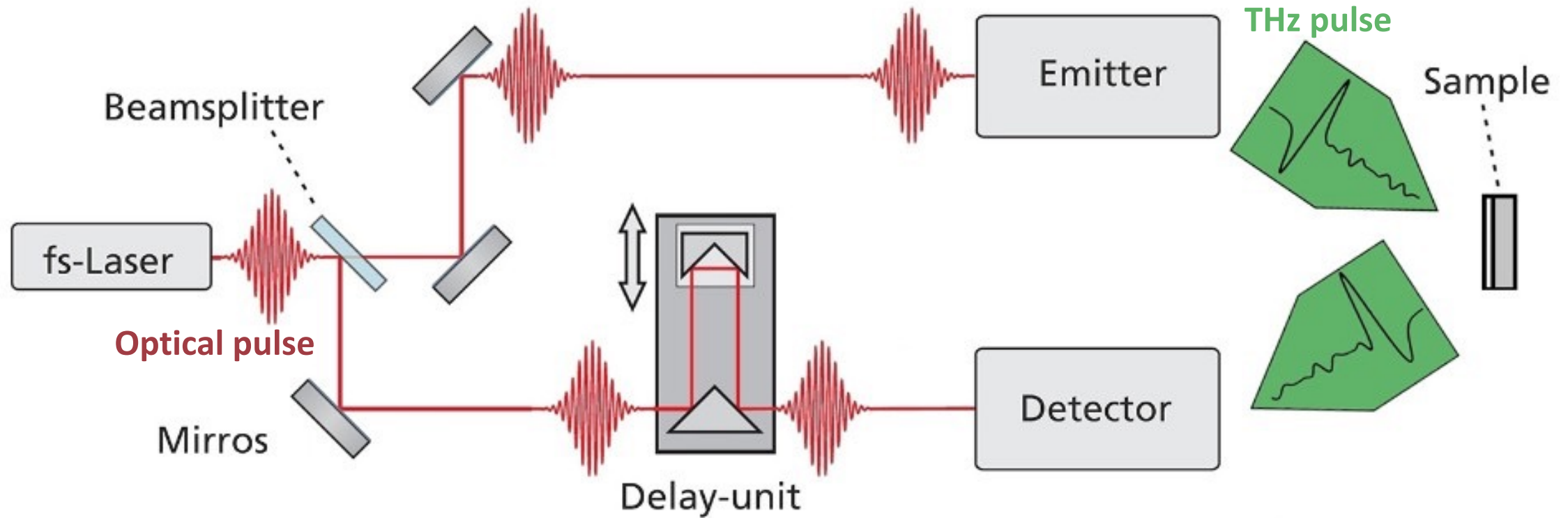
ω ... frequency components

Field-Resolved Detection in Non-linear THz Spectroscopy

- 1.) **Photoconductive antenna:** ✓ (limited to very high fields)
- 2.) **Bolometer:** ✗ (measures only intensity, slow response)
- 3.) **Electro-optical sampling:** ✓✓ (coherent detection)
 - **Crystal:** ZnTe or LiTaO₃ (nonlinear)
 - **Pockels effect:** THz field changes the refractive index
 - **Field-induced birefringence:** change of optical probe polarization



THz Time-Domain Spectroscopy (TDS) Setup

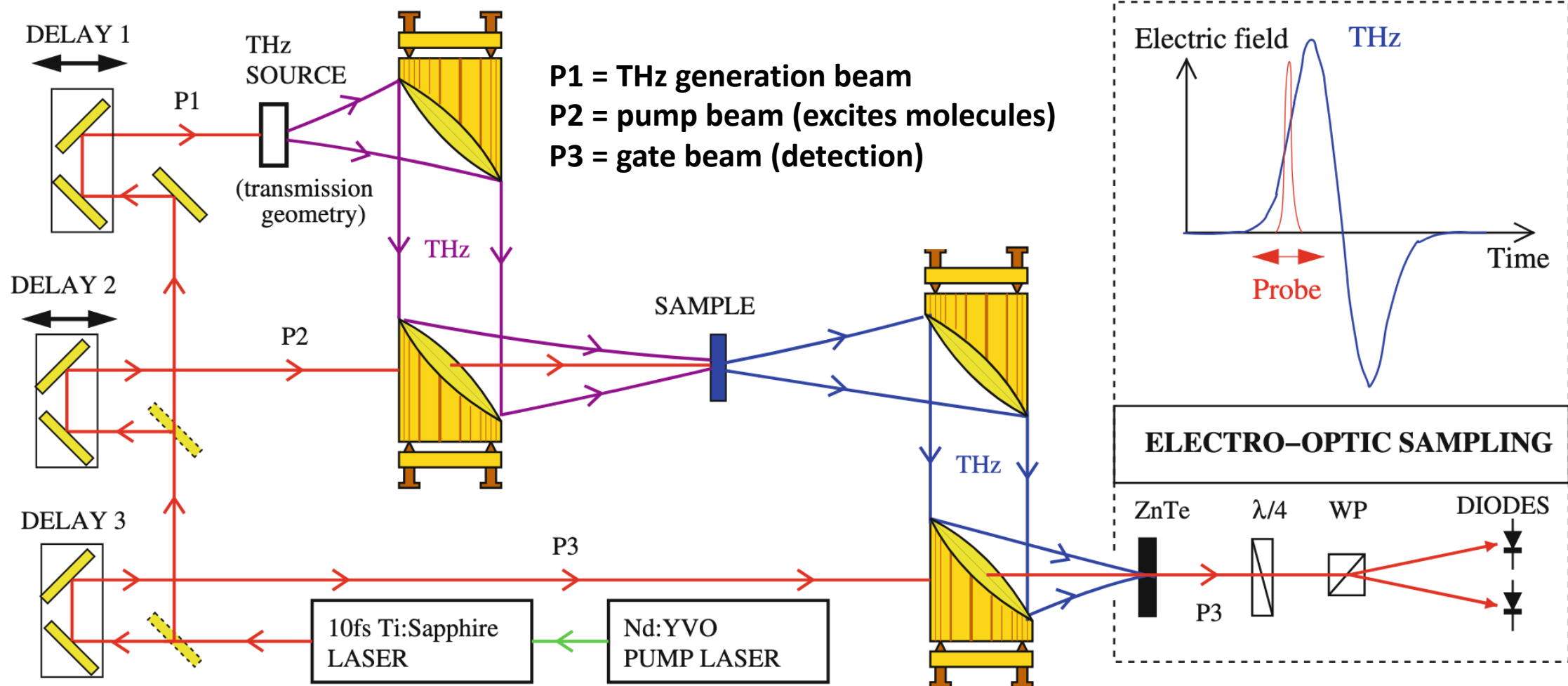


THz-TDS ≠ time-resolved spectroscopy!

Here, the “time domain” refers to measuring the electric THz field as a function of delay time.

Picture taken from: <https://www.itwm.fraunhofer.de/en/departments/processes-materials/materials-characterization-services-products-competences/terahertz-spectroscopy.html>

Optical-Pump-THz-Probe (OPTP) Spectroscopy Setup



Example: Catching an Electron Before it Becomes Solvated

Main idea:

When iodide in H₂O absorbs light, an electron is ejected.

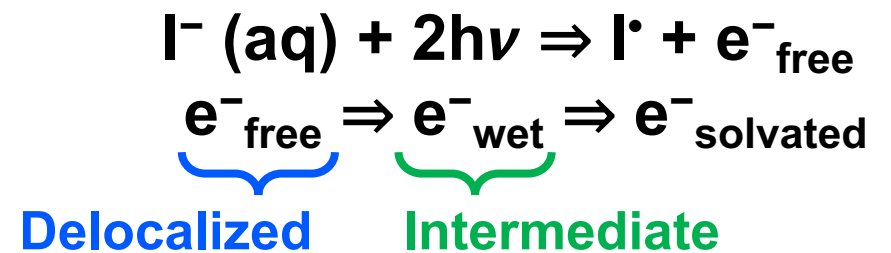
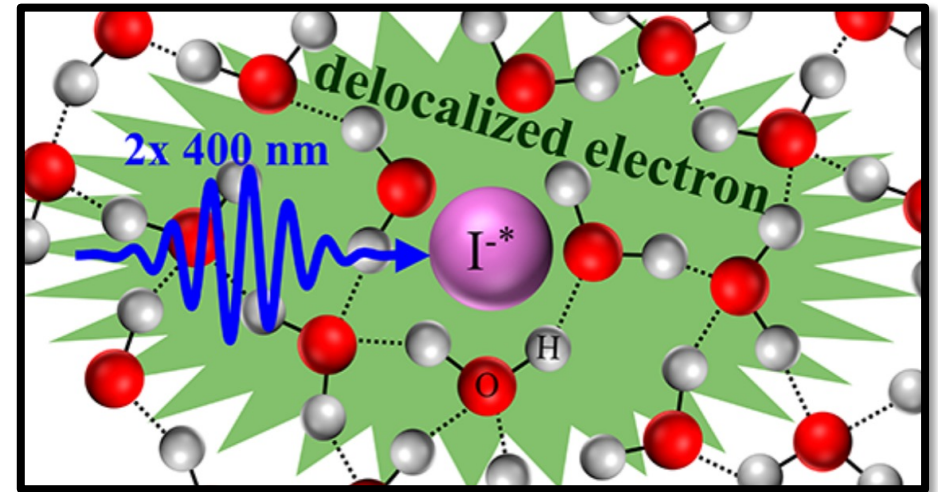
→ What happens during the first few 100 fs?

Why is this interesting?

Before becoming a fully solvated electron, the charge may exist in a short-lived, highly delocalized state.

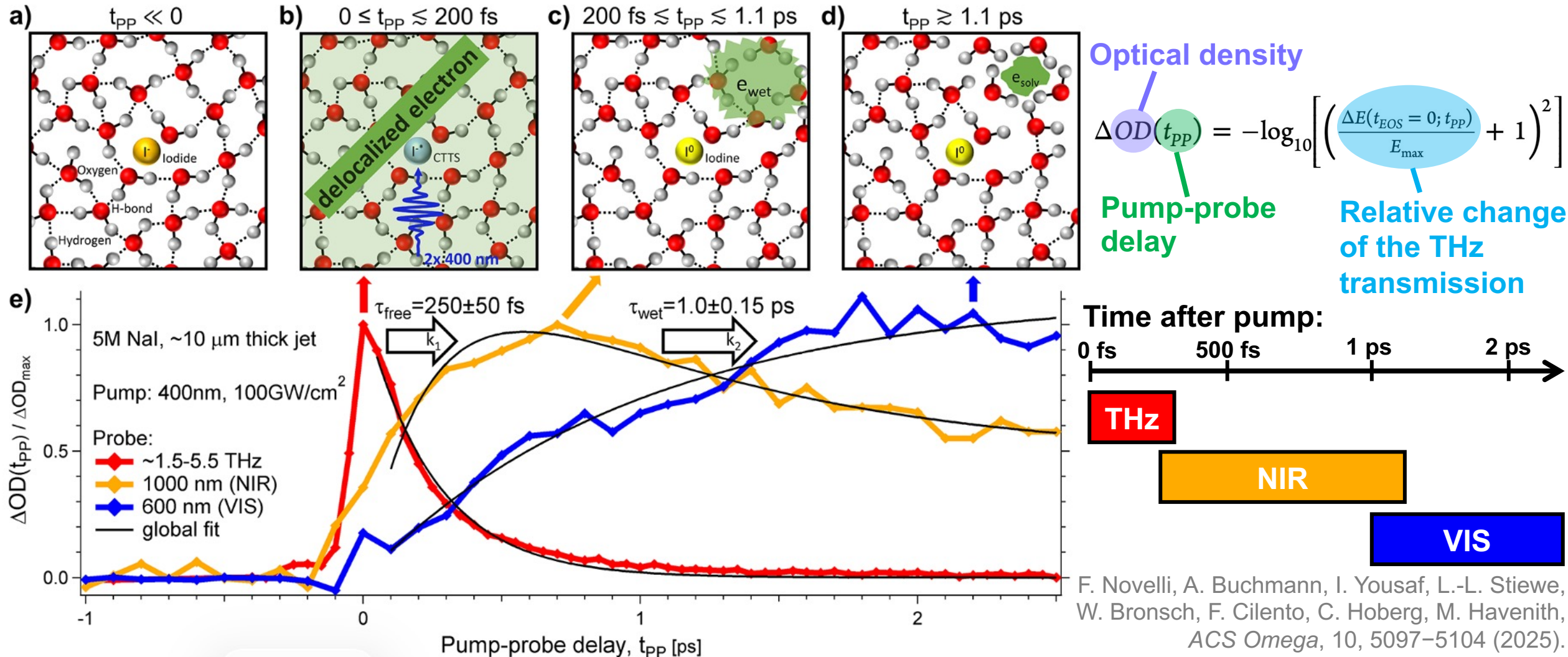
Why is OPTP suitable?

It captures the **earliest electronic dynamics!**



F. Novelli, A. Buchmann, I. Yousaf, L.-L. Stiewe, W. Bronsch, F. Cilento, C. Hoberg, M. Havenith, *ACS Omega*, 10, 5097–5104 (2025).

Example: Catching an Electron Before it Becomes Solvated



F. Novelli, A. Buchmann, I. Yousaf, L.-L. Stiewe, W. Bronsch, F. Cilento, C. Hoberg, M. Havenith, ACS Omega, 10, 5097–5104 (2025).

3.) Nonlinear THz Spectroscopy:

- 3.a.) From probing to driving: entering the nonlinear regime
- 3.b.) Strong-field THz sources
- 3.c.) Field-resolved THz detection
- 3.d.) THz time-domain spectroscopy (THz-TDS)
- 3.e.) Optical-pump-THz-probe spectroscopy and examples

4.) Summary:

- 4.a.) Take-home messages
- 4.b.) Questions for self-studying
- 4.c.) Further reading

Summary & Take-Home Messages

The THz gap is **no longer an inaccessible gap**.

THz spectroscopy provides access to **low-energy excitations** that are difficult to observe with many other spectroscopic techniques.

The key strength of THz spectroscopy is its **sensitivity to collective motions, intermolecular interactions and dynamics on picosecond timescales**.

THz spectroscopy is **rarely an isolated technique**.

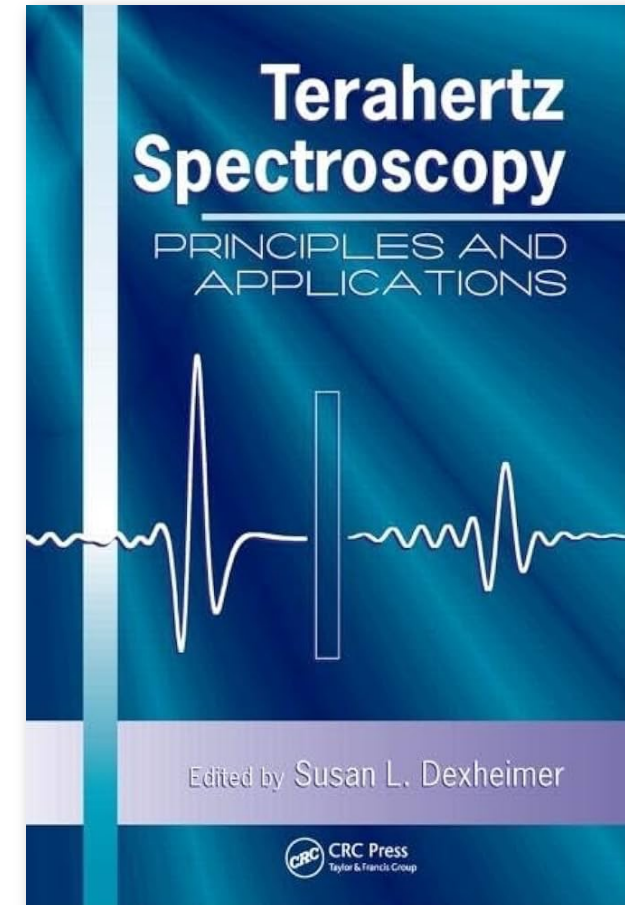
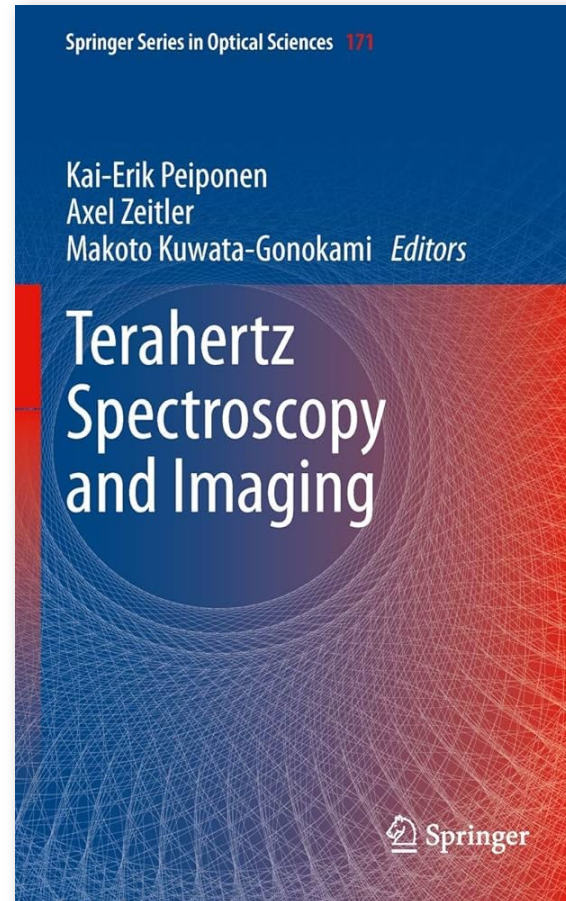
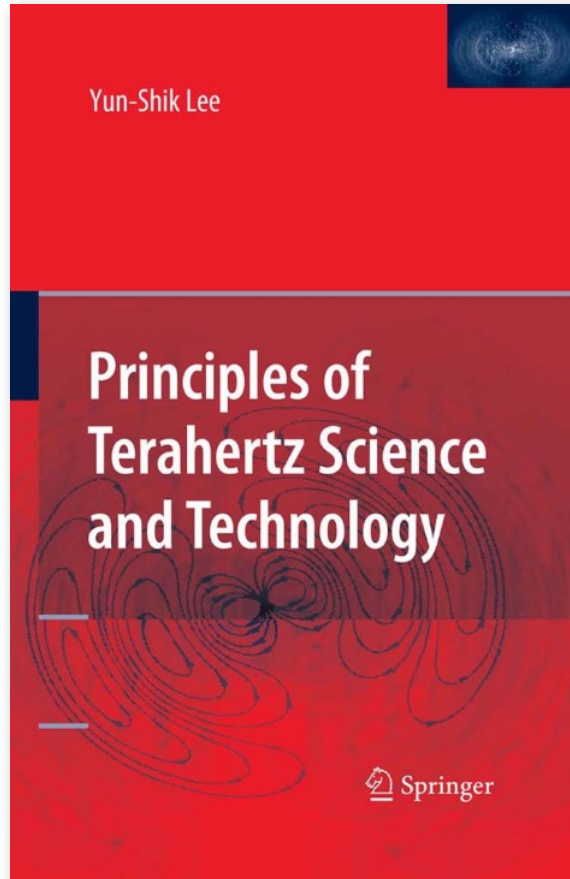
So whenever you hear “THz spectroscopy”, the first question should be: Is the THz field acting as a gentle **probe**, or is it **strong** enough to drive the sample? This distinction is central to understanding **linear vs. nonlinear THz spectroscopy**.

Guiding Questions for Reading a THz Spectroscopy Paper

- 1.) What is the scientific question?
- 2.) Why is THz spectroscopy suitable for this question?
3. What do we learn that would be difficult to see without THz spectroscopy?
- 3.) What experimental approach is used (FTIR, THz-TDS, OPTP)?
- 4.) Is the experiment linear or nonlinear?
- 5.) What is actually measured?
- 6.) What is used as a reference or background?
- 7.) How are the spectral features interpreted?
- 8.) What are the limitations of the experiment?
- 9.) What is the main take-home message?



Further Reading



Most figures in this lecture were either self-created or adapted from the book recommended at the beginning of the lecture. A few additional figures were adapted from the books shown on this slide.